DOUGLAS COUNTY GOVERNMENT

Finance Department ~ Purchasing Division
100 Third Street, Suite 130
Castle Rock, Colorado 80104
Telephone: 303-660-7434
www.douglas.co.us

INVITATION FOR BID (IFB)
NO. 041-21
HVAC FILTERS and BELTS

YOUR BID RESPONSE MUST BE RECEIVED NO LATER THAN NOVEMBER 30, 2021 @ 11:00AM

BIDDER'S CERTIFICATION

We offer to furnish to Douglas County the materials, supplies, products and/or services requested in accordance with the specifications and subject to the terms and conditions of the purchase(s) described herein:

BIDDER: AIR FILTER SOLUTIONS INC.

ADDRESS: 2500 W 8TH AVE, UNIT B

CITY: DENVER **STATE:** CO **ZIP:** 80204

TELEPHONE NUMBER: 303-893-3030 **FAX NUMBER:** 303-893-3031

EMAIL: mburt@airfiltersolutionsinc.com

BY: Mike Burt

(Printed or Typed Name)

Mike Burt Digitally signed by Mike Burt DN: cn=Mike Burt, o=Air Filter Solutions Inc., ou=Sales, email=mburt@airfiltersolutions.com, c=US Date: 2021.11.24 13:55:35 -07'00'

(Written Signature)

TAXPAYER I.D. NUMBER: 20-5110463

Signature constitutes acceptance of all terms and conditions listed on this form and all documents attached.

Please submit five (5) copies of your bid response in a sealed envelope that is clearly marked with the Invitation for Bid (IFB) information listed above. Mail or hand carry all bid responses to Douglas County Government, Finance Department, Purchasing Division, 100 Third Street, Suite 130, Castle Rock, Colorado 80104, prior to the due date and time. Electronic and/or faxed bid responses will not be accepted. It is the sole responsibility of the bidder to see that their bid response is received on time. Bids will not be considered which are received after the time stated, and any bids so received will be returned unopened. If closure of Douglas County buildings occurs on the day of a bid opening, the bid response must be delivered to the Purchasing Division before 12 noon on the following business day.

Douglas County Government reserves the right to reject any or all bids, to waive formalities, informalities, or irregularities contained in a said bid and furthermore, to award a contract for items herein, either in whole or in part, if it is deemed to be in the best interest of the County to do so. Additionally, we reserve the right to negotiate optional items and/or services with the successful bidder.

AIRFI-1 OP ID: CC

CERTIFICATE OF LIABILITY INSURANCE

DATE (MM/DD/YYYY) 06/17/2021

THIS CERTIFICATE IS ISSUED AS A MATTER OF INFORMATION ONLY AND CONFERS NO RIGHTS UPON THE CERTIFICATE HOLDER. THIS CERTIFICATE DOES NOT AFFIRMATIVELY OR NEGATIVELY AMEND, EXTEND OR ALTER THE COVERAGE AFFORDED BY THE POLICIES BELOW. THIS CERTIFICATE OF INSURANCE DOES NOT CONSTITUTE A CONTRACT BETWEEN THE ISSUING INSURER(S), AUTHORIZED REPRESENTATIVE OR PRODUCER, AND THE CERTIFICATE HOLDER.

IMPORTANT: If the certificate holder is an ADDITIONAL INSURED, the policy(ies) must be endorsed. If SUBROGATION IS WAIVED, subject to the terms and conditions of the policy, certain policies may require an endorsement. A statement on this certificate does not confer rights to the certificate holder in lieu of such endorsement(s).

PRODUCER Colling Insurance Services Inc 777 S. Wadsworth Blvd., #1-100 Lakewood, CO 80226 Bradford W. Colling	CONTACT NAME: Cindy Cole PHONE (A/C, No, Ext): 303-987-3331 FAX (A/C, No): 303-987-2881 E-MAIL ADDRESS: ccole@collinginsurance.com
INSURED Air Filter Solutions, Inc. 2500 W. 8th Ave. Suite B Denver, CO 80204	INSURER(S) AFFORDING COVERAGE — NAIC # INSURER A : Pinnacol Assurance — 41190 INSURER B : Nationwide Insurance INSURER C : Owners Insurance Company — 18988 INSURER D : INSURER E : INSURER F :

COVERAGES **CERTIFICATE NUMBER:** **REVISION NUMBER:**

THIS IS TO CERTIFY THAT THE POLICIES OF INSURANCE LISTED BELOW HAVE BEEN ISSUED TO THE INSURED NAMED ABOVE FOR THE POLICY PERIOD INDICATED. NOTWITHSTANDING ANY REQUIREMENT, TERM OR CONDITION OF ANY CONTRACT OR OTHER DOCUMENT WITH RESPECT TO WHICH THIS CERTIFICATE MAY BE ISSUED OR MAY PERTAIN, THE INSURANCE AFFORDED BY THE POLICIES DESCRIBED HEREIN IS SUBJECT TO ALL THE TERMS, EXCLUSIONS AND CONDITIONS OF SUCH POLICIES. LIMITS SHOWN MAY HAVE BEEN REDUCED BY PAID CLAIMS.

INSR LTR	TYPE OF INSURANCE	ADDL INSD	SUBR WVD	POLICY NUMBER	POLICY EFF (MM/DD/YYYY)	POLICY EXP (MM/DD/YYYY)	LIMITS	
B	X COMMERCIAL GENERAL LIABILITY						EACH OCCURRENCE	$ 1,000,000
	CLAIMS-MADE X OCCUR	X		ACPBPWD7522286092	07/07/2021	07/07/2022	DAMAGE TO RENTED PREMISES (Ea occurrence)	$ 300,000
	Business Owners						MED EXP (Any one person)	$ 5,000
							PERSONAL & ADV INJURY	$ 1,000,000
	GEN'L AGGREGATE LIMIT APPLIES PER:						GENERAL AGGREGATE	$ 2,000,000
	X POLICY PRO-JECT LOC						PRODUCTS - COMP/OP AGG	$ 2,000,000
	OTHER:							$
C	AUTOMOBILE LIABILITY						COMBINED SINGLE LIMIT (Ea accident)	$ 1,000,000
	X ANY AUTO			5184683600	07/07/2021	07/07/2022	BODILY INJURY (Per person)	$
	ALL OWNED AUTOS / SCHEDULED AUTOS						BODILY INJURY (Per accident)	$
	HIRED AUTOS / NON-OWNED AUTOS						PROPERTY DAMAGE (Per accident)	$
								$
	UMBRELLA LIAB OCCUR						EACH OCCURRENCE	$
	EXCESS LIAB CLAIMS-MADE						AGGREGATE	$
	DED RETENTION $							$
A	WORKERS COMPENSATION AND EMPLOYERS' LIABILITY Y / N						X PER STATUTE OTH-ER	
	ANY PROPRIETOR/PARTNER/EXECUTIVE OFFICER/MEMBER EXCLUDED? Y	N / A		4107099	08/01/2021	08/01/2022	E.L. EACH ACCIDENT	$ 1,000,000
	(Mandatory in NH) If yes, describe under DESCRIPTION OF OPERATIONS below						E.L. DISEASE - EA EMPLOYEE	$ 1,000,000
							E.L. DISEASE - POLICY LIMIT	$ 1,000,000

DESCRIPTION OF OPERATIONS / LOCATIONS / VEHICLES (ACORD 101, Additional Remarks Schedule, may be attached if more space is required)

Douglas County Facilities is included as Additional Insured under General Liability per form PB0448 1114 attached.

CERTIFICATE HOLDER

Douglas County Facilities
3026 N. Industrial Way
Castle Rock, CO 80109

CANCELLATION

SHOULD ANY OF THE ABOVE DESCRIBED POLICIES BE CANCELLED BEFORE THE EXPIRATION DATE THEREOF, NOTICE WILL BE DELIVERED IN ACCORDANCE WITH THE POLICY PROVISIONS.

AUTHORIZED REPRESENTATIVE

Cynthia J. Cole

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Air Filter Solutions Inc.
2500 WEST 8TH AVENUE, UNIT B
DENVER, CO 80204-3752
Phone: 303-893-3030 Fax: 303-893-3031

November 29, 2021

From: Mike Burt
IFB: NO. 041-21

BID SPECIFICATIONS FOR PURCHASE OF HVAC FILTERS and BELTS

6) References

- CITY OF LONGMONT FACILITIES DEPARTMENT
  1100 South Sherman St., Longmont, CO 80501
  Contact: Jeff Seader, Facilities Manager
  Phone: 303774-4531
  Email: jeff.seader@longmontcolorado.gov

- CONTROL SERVICE CENTER, INC.
  3280 Pearl St., Boulder, CO 80301
  Contact: Brandon Strouse or Kyle Strouse, owners/general manager
  Phone: 303-4497405
  Email: brandon@controlservicecenter.com , kyle@controlservicecenter.gov

- CITY OF THORNTON
  9500 Civic Center Drive, Thornton, CO 80229
  Contact: Jim Tully, Facilities Manager
  Phone: 303-7460763
  Email: jim.tully@cityofthorton.net

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC MERV 9 PLEAT FILTERS
BID SCHEDULE

TYPE: HVAC MERV 9 PLEAT FILTERS						
Size	**Estimated Annual Usage**	**Minimum Order Quantity**	**Volume Price Break**	**Unit Price**	**Manufacturer**	**Model Number**
16x16x2	8	1	none	12.78	Camfil	16162-D9
20x16x2	295	1	none	12.62	Camfil	20162-D9
16x25x2	162	1	none	15.24	Camfil	16252-D9
20x20x2	245	1	none	14.53	Camfil	20202-D9
20x24x2	133	1	none	17.04	Camfil	20242-D9
20x25x2	265	1	none	18.25	Camfil	20252-D9
25x16x2	245	1	none	15.24	Camfil	25162-D9
25x20x2	356	1	none	17.72	Camfil	25202-D9
25x25x2	10	1	none	22.75	Camfil	25252-D9
24x12x2	36	1	none	12.07	Camfil	24122-D9
24x16x2	115	1	none	14.55	Camfil	24162-D9
24x20x2	118	1	none	16.53	Camfil	24202-D9
24x24x2	56	1	none	19.88	Camfil	24242-D9
13.5x19.5x2	4	1	none	16.16	Camfil	19H13H2-D9
34.5x15.5x2	2	1	none	28.74	Camfil	34H15H2-D9
34.5x24.5x2	2	1	none	40.00	Camfil	34H24H2-D9
20x16x4	16	1	none	24.35	Camfil	20164-D9
20x20x4	12	1	none	25.79	Camfil	20204-D9
24x20x4	32	1	none	32.62	Camfil	24204-D9
24x12x4	12	1	none	23.14	Camfil	24124-D9
24x24x4	40	1	none	33.87	Camfil	24244-D9
25x16x4	66	1	none	28.24	Camfil	25164-D9
25x20x4	178	1	none	34.56	Camfil	25204-D9

Performance Criteria	
ECI Value of 5 Stars	YES / NO
Initial Resistance (in W.G.)	.30"
Final Resistance (in W.G.)	1.00"
Minimum Composite Average Efficiency - % E1	8
Minimum Composite Average Efficiency - % E2	39
Minimum Composite Average Efficiency - % E3	80
Merv – Rating (Minimum 9A)	YES / NO
Integrity guaranteed to 2" W.G.	YES / NO
Underwriters Laboratories as UL Class 2	YES / NO

Note: UL has now switched over to UL 900 and Camfil qualifies on all filters

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC MERV 9 PLEAT FILTERS
BID SCHEDULE - Continued

Physical Characteristics		
Beverage Board - 28 Point high wet strength		YES / NO
Frame Bonded to Media on All Sides		YES / NO
Media Type (i.e., synthetic, cotton)		
Media Thickness (0.15")		YES / NO

COMMENTS:

Our supporting Data shall provide A product test report for this MERV 9 listed efficiency including all details as prescribed in ASHRAE Standard 52.2, including all details as prescribed in ASHRAE Standard 52.2 including Appendix J. All pages of the test report shall be included to substantiate appendix J value. This is of key importance.

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC RING PANEL 2 PLY
BID SCHEDULE

TYPE: HVAC RING PANEL 2 PLY						
Size	**Estimated Annual Usage**	**Minimum Order Quantity**	**Volume Price Break**	**Unit Price**	**Manufacturer**	**Model Number**
POLY WIRE FRAME						
11-3/4x47x1	4	12	none	$12.00	filberbond / AFS	500-599-W
9x24x1	12	12	none	$7.00	filberbond / AFS	0-299-W
12x13x1	120	12	none	$7.00	filberbond / AFS	0-299-W
12x7x1	12	12	none	$7.00	filberbond / AFS	0-299-W
15x13x1	12	12	none	$7.00	filberbond / AFS	0-299-W
15x17x1	120	12	none	$7.00	filberbond / AFS	0-299-W
19x30x1	72	12	none	$12.00	filberbond / AFS	500-599-W
19x19x1	24	12	none	$7.00	filberbond / AFS	300-399-W
11-3/4x54x1	4	12	none	$15.00	filberbond / AFS	600-699-W
POLY PANEL						
11-3/4x47x1	4	12	none	$12.00	filberbond / AFS	500-599-W

Performance Criteria	2 PLY
Test Air Flow Rate CFM	1200
Initial Resistance (in W.G.)	.15"
Final Resistance (in W.G.)	.50"
Merv – Rating 7 ASHRARE 52.2 – 1999	N/A
Anti-Microbial Biocide (NOT BIOSTATE)	YES/NO

Physical Characteristics	2 PLY	
Media Type	blue and white tackified & white dry	
Wire Frame Gauge	9 guage	
Filter Has Internal Stabilizer Seal	YES / NO	on common sizes
Heat Sealed Around Internal Wire Grid	YES / NO	on common sizes

COMMENTS:

In custom sizes we build as a stitched / sewn around wire gage frame and if in large quantites they can be internal stabilizer and or heat sealed. In past history, sometimes the tech may ask for white 1/2" thick media non-tac verses 1" blue and white tac. We have always been flexable to your needs.

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

MINI-PLEAT 4V-CELL TYPE
FILTER U.L. CLASS 2
BID SCHEDULE

TYPE: MINI-PLEAT 4V CELL TYPE FILTER						
Size	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer	Model Number
10P MERV 13A - HI-FLO-ES BAG						
20x24x22	4	2	none	89.49	CAMFIL	242022-8-MV13ES
24x24x22	24	2	none	108.79	CAMFIL	242422-10-MV13ES
10P MERV 14A - HI-FLO-ES BAG						
12x24x22	8	4	none	60.89	CAMFIL	241222-5-MV14ES
20x24x22	19	2	none	93.54		242022-8-MV14ES
DF MERV 13A – 4/5-V MINI-PLEAT						
24x12x12	13	1	none	129.72	CAMFIL	241212-DFMV13
20x20x12	52	1	none	202.74	CAMFIL	202012-DFMV13
24x20x12	112	1	none	202.74	CAMFIL	242012-DFMV13
24x24x12	72	1	none	221.78	CAMFIL	242412-DFMV13
DF MERV 14A – 4/5-V MINI-PLEAT						
24x12x12	12	1	none	139.07	CAMFIL	241212-DFMV14
24x24x12	16	1	none	229.29	CAMFIL	242412-DFMV14

COMMENTS:

Our supporting Data shall provide product test reports for each listed efficiency including all details as prescribed in ASHRAE Standard 52.2, including all details as prescribed in ASHRAE Standard 52.2 including Appendix J. All pages of the test report shall be included to substantiate appendix J value. This is of key importance.

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
3L180	8			6.99	GATES
3L190	6			6.99	
3L220	4			6.99	
3L270	1			6.99	
3L280	2			7.18	
3L300	3			7.18	
3L370	1			7.18	
3L380	1			7.85	
3L390	6			7.85	
3VX300	2			15.83	
3VX530	4			22.76	
3VX600	14			24.61	
3VX670	8			26.53	
3VX900	4			36.36	
4L200	12			7.45	
4L210	10			7.45	
4L220	2			7.45	
4L230	1			7.45	
4L240	1			7.45	
4L250	4			7.45	
4L290	3			7.98	
4L300	9			8.07	
4L310	4			8.24	
4L320	3			8.33	
4L420	1			9.98	
4L590	1			12.26	
4L670	2			13.41	
4L800	8			15.81	

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE - Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
5VX1000	12			88.59	GATES
5VX1030	4			91.25	
5VX1120	8			99.41	
5VX450	2			40.00	
5VX490	2			40.00	
5VX510	6			45.18	
5VX530	2			46.12	
5VX550	2			47.86	
5VX630	4			54.73	
5VX660	2			57.51	
5VX670	6			58.33	
5VX710	2			61.93	
5VX780	2			68.89	
5VX900	2			78.14	
5VX960	4			86.28	
A23	2			8.36	
A26	3			8.36	
A29	1			8.99	
A32	8			9.49	
A33	8			9.58	
A37	4			10.42	
A38	5			10.59	
A39	3			10.82	
A41	4			11.16	
A42	1			11.41	
A44	3			11.73	
A45	7			11.87	
A48	3			12.19	
A49	3			12.34	
A50	1			12.46	
A51	3			12.59	

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
A55	11			13.18	GATES
A60	4			13.94	
A61	8			14.15	
A65	2			14.76	
A92	2			21.95	
AX21	12			9.68	
AX22	11			10.13	
AX23	12			10.59	
AX24	2			10.76	
AX25	1			10.91	
AX26	4			11.33	
AX27	2			11.54	
AX29	4			11.92	
AX30	6			11.79	
AX31	2			12.19	
AX32	2			12.44	
AX34	2			12.91	
AX35	2			13.07	
AX36	4			13.55	
AX37	4			13.96	
AX38	4			14.11	
AX39	2			14.32	
AX40	2			14.53	
AX41	4			14.84	
AX42	17			14.99	
AX45	2			15.56	
AX51	4			16.57	
AX53	8			16.91	
AX54	4			17.81	
AX55	8			17.46	
AX56	8			17.54	

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
AX58	2			10.42	GATES
AX59	2			18.33	
AX61	4			18.75	
AX62	4			18.87	
AX64	8			19.32	
AX67	2			20.02	
AX68	12			20.19	
AX69	6			20.64	
AX70	4			21.04	
AX71	8			21.25	
AX96	8			28.96	
B112	4			36.07	
B70	6			22.64	
B73	3			23.44	
B78	2			25.04	
B92	6			29.58	
B93	4			30.02	
BX103	6			43.01	
BX35	2			17.55	
BX38	18			19.32	
BX40	4			20.19	
BX41	2			20.64	
BX42	8			21.04	
BX45	2			22.56	
BX46	1			22.82	
BX47	2			23.01	
BX48	4			23.53	
BX50	4			24.51	
BX59	4			26.16	
BX61	6			26.68	
BX62	2			27.04	
BX63	3			27.39	↓

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

HVAC BELTS
BID SCHEDULE – Continued

Model Number	Estimated Annual Usage	Minimum Order Quantity	Volume Price Break	Unit Price	Manufacturer
BX66	2			28.42	GATES
BX68	2			28.96	
BX70	7			29.74	
BX71	2			30.09	
BX74	3			31.39	
BX76	2			32.00	
BX77	2			32.49	
BX78	2			32.89	
BX80	9			33.79	
BX81	2			34.21	
BX82	2			35.08	
BX85	3			35.98	
BX93	2			39.48	
BX95	2			40.38	
M411425P	2				

COMMENTS:

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS

CELdek – Direct Evaporative Cooling/Humidifier Media
BID SCHEDULE

TYPE: CELdek – Direct Evaporative Cooling/Humidifier Media			
Estimated Usage	**Unit Price Per Cubic Foot**	**Manufacturer**	**Model Number**
214 cubic ft	**24.60**	**Munters Celdek**	**CD-CELDEK (NEED SIZE)**

COMMENTS:

THERE IS NO FORMALDEHYDE MANUFACTURED WITHIN THIS PRODUCT

Made in USA

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS
DETAILED BID SPECIFICATIONS

SECTION ONE
MERV-9 PLEATED PANEL FILTERS

1) SCOPE

This specification covers pleated panel filters that are a component of air-conditioning, heating, and ventilating systems.

2) PURPOSE

The purpose of this specification is to establish performance criteria and identify physical properties that are pertinent and necessary for proper filter performance. Conformance to all items in the specification is the responsibility of the bidder.

3) BID ENCLOSURES

The following items are required as part of the bid response.

3.1 A product sample for each distinctively different product design.

4) GENERAL

4.1 Air filters shall be medium efficiency ASHRAE pleated panels consisting of cotton and synthetic media, welded wire media support grid, and beverage board enclosing frame.

4.2 Sizes shall be noted on drawings or other supporting materials.

5) CONSTRUCTION

5.1 Filter media shall be two layers of polyester, lofted to a uniform depth of 0.11", and formed into a uniform radial pleat. Filter media shall have a MERV 9 efficiency after conditioning methods as stated in ASHRAE 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size, Appendix J or in the discharging method designated in ISO 16890.

5.2 A welded wire grid of 0.020" in diameter, spot-welded on one-inch centers and treated for corrosion resistance shall be bonded to the downstream side of the media to maintain radial pleats and prevent media oscillation.

5.3 An enclosing frame of no less than 28-point high wet-strength beverage board shall provide a rigid and durable enclosure. The frame shall be bonded to the media on all sides to prevent air bypass. Integral diagonal support members on the air entering and air exiting side shall be bonded to the apex of each pleat to maintain uniform pleat spacing in varying airflows.

6) PERFORMANCE

6.1 The filter shall have a Minimum Efficiency Reporting Value of MERV 9 when evaluated under the guidelines of ASHRAE Standard 52.2. It shall have a MERV-A of 9 when tested per Appendix J of the same standard. The media shall maintain or increase in efficiency over the life of the filter.

6.2 The filter shall have a minimum efficiency rating of ISO ePM10 -55 when tested to ISO standard 16890.

6.3 Initial resistance to airflow shall not exceed 0.30" w.g. for 2" configurations at 500 fpm, and 0.27" w.g. for 4" configurations at 500 fpm.

6.4 The filter shall have a Minimum Efficiency Reporting Value of MERV 9 when evaluated under the guidelines of ASHRAE Standard 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size, Appendix J. The media shall maintain or increase in efficiency over the life of the filter.

6.5 Initial resistance to airflow shall not exceed 0.32" w.g. for 2" configurations at 500 fpm, and 0.29" w.g. for 4" configurations at 500 fpm.

6.6 The filter shall have an Energy Cost Index (ECI) value of five stars.

6.7 The filter shall be classified by Underwriters Laboratories as UL Class 900.

6.8 Manufacturer shall provide evidence of facility certification to ISO 9001:2008.

6.9 Manufacturer shall guarantee the integrity of the filter pack to 2.0" w.g.

6.10 Manufacturer shall guarantee the efficiency level will always be at or exceed MERV 9 when tested according to ASHRAE 52.2, or ISO ePM_{10} 50% when tested to ISO 16890, throughout the entire service life of the filter.

6.11 Manufacturer shall guarantee the filter will last twelve (12) calendar months in applications operating at or less than 100 hours per week, before reaching 1.0" w.g. of static pressure across the filter OR the manufacturer shall guarantee the filter will last nine (9) calendar months in applications operating above 100 hours per week, before reaching 1.0" w.g. of static pressure across the filter. Both guarantees are to be valid at or below the rated air velocity of 500 fpm.

6.12 Supporting Data - Provide product test report including all details as prescribed in ASHRAE Standards 52.2, including Appendix J.

7) PACKAGING AND IDENTIFICATION

7.1 Air filter identification markings on each filter frame shall include
- 7.1.1.1 Filter Manufacture
- 7.1.1.2 Filter Type/Model Number
- 7.1.1.3 Nominal Filter Size/Actual Filter Size
- 7.1.1.4 MERV Rating
- 7.1.1.5 UL 900 Classification
- 7.1.1.6 Code Build Date

7.2 Filter packing carton markings shall include
- 7.2.1.1 Filter Manufacture
- 7.2.1.2 Filter Type/Model Number
- 7.2.1.3 Nominal Filter Size
- 7.2.1.4 MERV Rating
- 7.2.1.5 UL 900 Classification
- 7.2.1.6 Manufacture Part Number
- 7.2.1.7 Carton Quantity

7.3 Packaging shall provide adequate protection against damage or deterioration during shipment and allow complete identification of both the filters and the shipping container.

8) PERFORMANCE CRITERIA

8.1 The manufacturer shall have representative samples of the product requested in the bid solicitation tested by an Independent test laboratory approved by the purchaser. The test laboratory shall test one (1) representative filter, tested to ASHRAE standard 52.2.-2012 and with Appendix J, five (5) years or newer.

8.2 The resulting performance data shall be used to meet the following:

8.2.1 NOMINAL FILTER SIZE (W X H X D) -- 24x24x2
Gross Media Area -- 17.3

8.2.2 TEST CONDITIONS
Filter Face Velocity (CFM) -- 2000
Initial Clean Resistance (Inches W.G.) -- 0.32"
Final Dirty Resistance (Inches W.G.) -- 1.00"

8.2.3 PERFORMANCE REQUIREMENTS-ASHRAE 52.2-2012
Minimum Composite Average Efficiency (%) under Appendix J: E1: 8; E2: 39; E3: 80
Minimum Efficiency Reporting Value (MERV-A) 9

8.2.4 PERFORMANCE REQUIREMENTS-ASHRAE 52.-2012
Minimum Average Arrestance (%) -- 92
Measured dust holding capacity (DHC) -- 200 grams

8.3 The Supplier shall agree to have a representative make on-site inspections on a quarterly basis. The purpose of this inspection is to ensure that the filters are functioning properly.

9) BID ENCLOSURES

The following items are required as part of the bid response.

9.1 For each indicated filter size and capacity, one (1) laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012. Permission shall be granted by the bidder to verbally verify the performance results with the laboratory that performed the tests.

9.2 A product sample for each distinctively different product design.

camfil

30/30® Dual 9

High-Capacity MERV 9/9A Pleated Panel Filter

Camfil has reinvented an entire product category

Only MERV 9A Panel Filter Guaranteed to Last 9 - 12 Months

CLEAN AIR SOLUTIONS

camfil

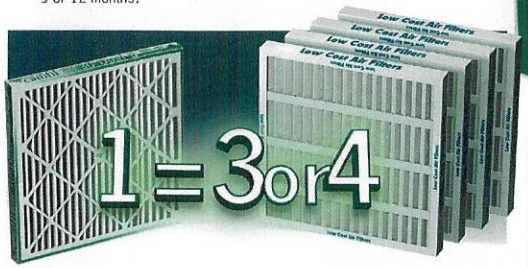
30/30® Dual 9

High-Capacity MERV 9/9A Pleated Panel Filter

DUAL 9 SETS A NEW STANDARD IN THE AIR FILTRATION INDUSTRY. This is a Game Changer.

What is Dual 9?

Dual 9 is a breakthrough product where every component has been engineered to deliver the highest level of continuous particle removal with the lowest energy consumption and longest service life. The Dual 9 comes with dual but equal MERV ratings (MERV 9 and MERV 9A), a dual layered gradient density media and dual performance guarantee options of 9 or 12 months.



For every Dual 9 filter, it will take three or four low cost pleated filters — waste disposal, landfill impact and transaction costs. At the same time significant reduction in energy consumption per year up to 25% per filter opening while having MERV9A efficiency (per ASHRAE standard 52.2 Appendix J).

Camfil, the First Innovator is still the Only Innovator.

Camfil asked users what they wanted in pleated panel filters and three objectives stood out:

1. Longest possible service life to reduce the need for frequent filter changes thus saving labor resources for more critical maintenance projects.
2. Reduced energy consumption in line with corporate cost saving and sustainability objectives.
3. Improved filtration for better protection of HVAC equipment — reduces the need for coil and system cleaning by 25%-50% compared to lower efficiency MERV 8 and MERV 7 solutions.

30/30® Dual 9 Pleat
High Lofted Dual Media
- Gradient density, mechanical filtration media

Competitors' V-Pleat
Coarse Fiber
- Utilizes electrostatically enhanced media
- Short service life, high energy consumption and particle capture degradation below rated efficiency.

Available in 2" and 4" Depths

Built Strong

Designed to withstand turbulent or harsh airflow that may be common in some HVAC systems.

Welded Steel Grid with Corrosion Inhibitor

A crossed wire grid, spot welded on one inch centers, provides industrial grade support to hold pleat formation throughout service life.

Proprietary Dual High Lofted Fibers

New media technology - Dual layered media combines two filtration layers in one package. Upstream white layer designed for larger particles and downstream green layer designed for finer particles. Combined solution provides a dust holding capacity 4 times higher than general commercial grade pleats.

Radial Pleats

Radial pleats expose all of the media area to the dirty airstream providing lower pressure drop and higher dirt holding capacity.

Diagonal Support

Diagonal support members bonded to the media maintain pleat separation and ensure filter rigidity.

Moisture Resistant Frame Board

28-point high wet-strength frame board is resistant to moisture and has a high tensile strength.

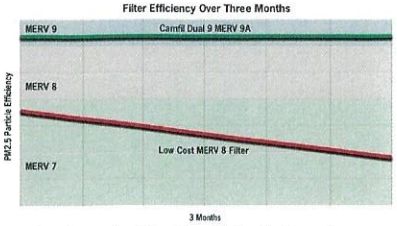
ENERGY COST INDEX

5-Star ECI Rating

Camfil's filters are rated as 5-Star filters through the Energy Cost Index (ECI) program. Based upon a five-star scale, the Energy Cost Index is an indicator of what a filter will cost over its lifetime. The best rating – five stars – indicates that the filter is the most energy-efficient, longest lasting filter available.

We Guarantee It!

Don't think it's too good to be true! Camfil was the first air filtration manufacturer to offer a written performance guarantee on product performance and now we have raised the bar to the highest level ever — providing our customers the highest "Total Cost of Ownership" value proposition with *no risk*.



Area between the efficiency lines reflects particulate passing through the filter, collecting on secondary filters, heating and cooling coils, all which hinder total ventilation operating efficiency.

Built to last. Tested for proof.

30/30® Dual 9 media is manufactured from a proprietary blend of fine fibers utilizing a mechanical particle capture principle.

The Dual 9 media was put through three years of real life testing to ensure performance under real life conditions and to establish the only guarantee of its kind. Highest energy saver, lifetime efficiency, and lasts 9 to 12 months...*Guaranteed!*

9 MONTHS — 30/30 DUAL 9 — 12 MONTHS

The electret charge of the competitors' coarse fiber filters dissipates over time resulting in a loss of efficiency long before what should be a full life cycle.

camfil

30/30® Dual 9

High-Capacity MERV 9/9A Pleated Panel Filter

Part Number	Description	Height (inches)	Width (inches)	Depth (inches)	Airflow Capacity (cfm)	Media Area (sq ft.)	Resistance at Capacity (inches of w.g.)	Pleats per Linear Foot
406331-001	30/30 D9 20X16X2	19.50	15.50	1.75	1110	9.9	0.30	15
406331-002	30/30 D9 20X20X2	19.50	19.50	1.75	1390	11.9	0.30	15
406331-003	30/30 D9 25X20X2	24.50	19.50	1.75	1740	14.9	0.30	15
406331-004	30/30 D9 25X16X2	24.50	15.50	1.75	1390	12.4	0.30	15
406331-005	30/30 D9 24X24X2	23.38	23.38	1.75	2000	17.3	0.30	15
406331-006	30/30 D9 24X12X2	23.38	11.38	1.75	1000	8.4	0.30	15
406331-007	30/30 D9 20X12X2	19.50	11.88	1.75	835	7.4	0.30	15
406331-008	30/30 D9 20X10X2	19.50	9.50	1.75	700	6.0	0.30	15
406331-009	30/30 D9 20X14X2	19.50	13.50	1.75	975	8.3	0.30	15
406331-010	30/30 D9 25X14X2	24.50	13.50	1.75	1220	10.4	0.30	15
406331-011	30/30 D9 20X15X2	19.50	14.50	1.75	1045	9.3	0.30	15
406331-012	30/30 D9 24X20X2	23.50	19.50	1.75	1670	14.3	0.30	15
406331-013	30/30 D9 20X18X2	19.50	17.50	1.75	1250	10.8	0.30	15
406331-014	30/30 D9 25X18X2	24.50	17.50	1.75	1565	13.5	0.30	15
406331-015	30/30 D9 24X18X2	23.50	17.50	1.75	1500	13.0	0.30	15
406331-016	30/30 D9 24X16X2	23.50	15.50	1.75	1335	11.8	0.30	15
406331-018	30/30 D9 25X25X2	24.50	24.50	1.75	2170	19.0	0.30	15
406331-019	30/30 D9 16X16X2	15.50	15.50	1.75	890	7.8	0.30	15
406331-021	30/30 D9 20X25X2	19.50	24.50	1.75	1740	14.9	0.30	15
406331-022	30/30 D9 16X25X2	15.50	24.50	1.75	1390	12.4	0.30	15
406331-023	30/30 D9 20X24X2	19.50	23.50	1.75	1670	14.3	0.30	15
406332-001	30/30 D9 20X16X4	19.50	15.50	3.75	1110	15.7	0.27	11
406332-002	30/30 D9 20X20X4	19.50	19.50	3.75	1390	18.9	0.27	11
406332-003	30/30 D9 25X20X4	24.50	19.50	3.75	1740	23.6	0.27	11
406332-004	30/30 D9 25X16X4	24.50	15.50	3.75	1390	19.7	0.27	11
406332-005	30/30 D9 24X24X4	23.38	23.38	3.75	2000	27.7	0.27	11
406332-006	30/30 D9 24X12X4	23.38	11.38	3.75	1000	13.9	0.27	11
406332-012	30/30 D9 24X20X4	23.50	19.50	3.75	1670	22.7	0.27	11
406332-015	30/30 D9 24X18X4	23.50	17.50	3.75	1500	20.2	0.27	11
406332-020	30/30 D9 25X29X4	24.38	28.38	3.75	2520	35.4	0.27	11
406332-021	30/30 D9 20X25X4	19.38	24.38	3.75	1740	23.5	0.27	11
406332-022	30/30 D9 16X25X4	15.38	24.38	3.75	1390	18.8	0.27	11
406332-023	30/30 D9 20X24X4	19.38	23.38	3.75	1670	22.6	0.27	11

Unprecedented Industry Guarantee

We are so confident in our filter's performance, we are the only manufacturer that GUARANTEES our filters will hold their rated efficiency for their entire service life, as well as last 9 or 12 months in service, based on the application. If our filters fail to hold a MERV 9 efficiency or fail to last 9 or 12 months in service, we'll replace them, FREE of charge.

Camfil GUARANTEES the 30/30 Dual 9 will last 9 months in heavy load applications and 12 months in light to moderate applications when operated at or below the rated airflow capacity for that filter.
For guarantee details, visit www.camfil.us.

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1 North Corporate Drive | Riverdale, NJ 07457
Phone: 973.616.7300 | Fax: 973.616.7771
camfil@camfil.com
www.camfil.us

www.camfil.us

For further information, please contact your nearest Camfil office.

Blue Heaven Technologies

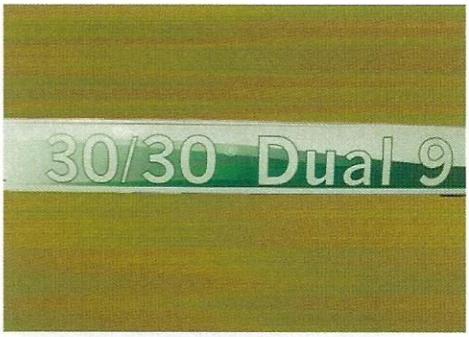
2820 S. English Station Road - Louisville, KY 40299
Tel: (502) 357-0132 Fax (502) 267-8379

Date: 14-Dec-16 **TEST NO.** 16-1893Rev1

Test Report
ANSI/ASHRAE Standard 52.2-2012 with 2015 Supplement And Appendix J Discharge

Filter Description

Manufacturer	Camfil
Filter Model	30/30 Dual 9
Part Number	406331005
Generic Filter Type	Pleated Prefilter
Nominal Dimensions (H x W x D)	24" x 24" x 2"
Pocket / Pleat Quantity	15 Pleats/Linear Foot
Media Type	Dual Layered Polyester
Est. Gross Media Area	17.3 Ft²
Adhesive Type	N/A



Test Conditions

Conditioning Aerosol Type	KCL		
Loading Dust Type	ASHRAE	Test Air Temp (degrees F.)	71
Barometric Pressure (In. Hg.)	29.59	Relative Humidity (%)	47

Test Results

Airflow Rate (CFM)	**1968**
Nominal Face Velocity (fpm)	**492**
Initial Resistance (in WG)	**0.28**
Final Resistance (in WG)	**1.00**
Dust Fed (gms) to Final Resistance	**216**
E1 (%) Composite Minimum Avg. Efficiency 0.30 - 1.0 um	**8**
E2 (%) Composite Minimum Avg. Efficiency 1.0 - 3.0 um	**39**
E3 (%) Composite Minimum Avg. Efficiency 3.0 - 10.0 um	**80**
Minimum Efficiency Reporting Value (MERV)	**MERV 9-A @ 1968 CFM**

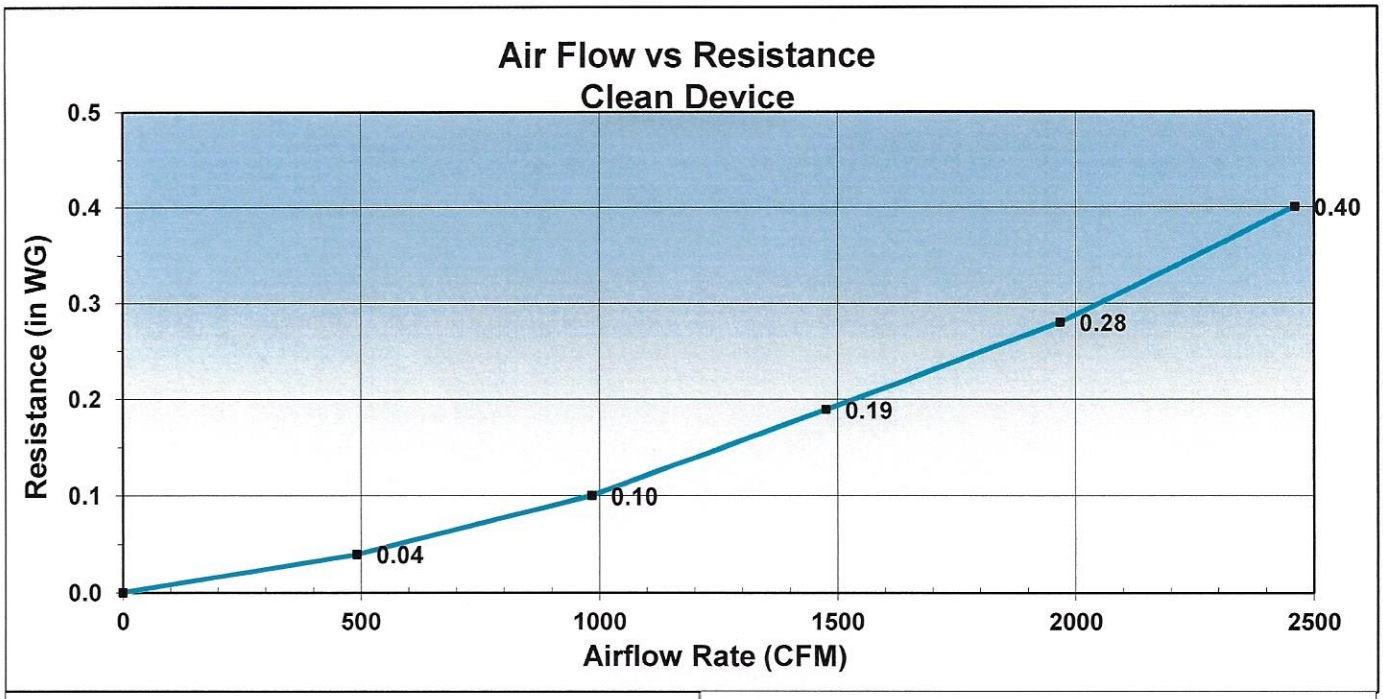
Comments Tested For: Camfil USA

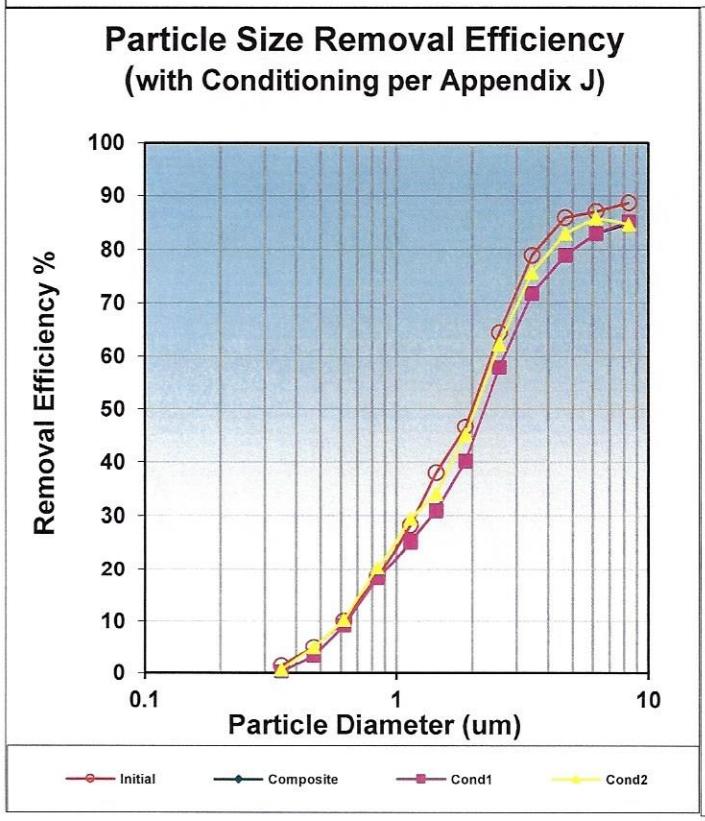
Final Pressure Drop ("w.c.)	**1.00"w.c.**
Dust Holding Capacity (gms)	**199**
Average Arrestance (%)	**92.0**

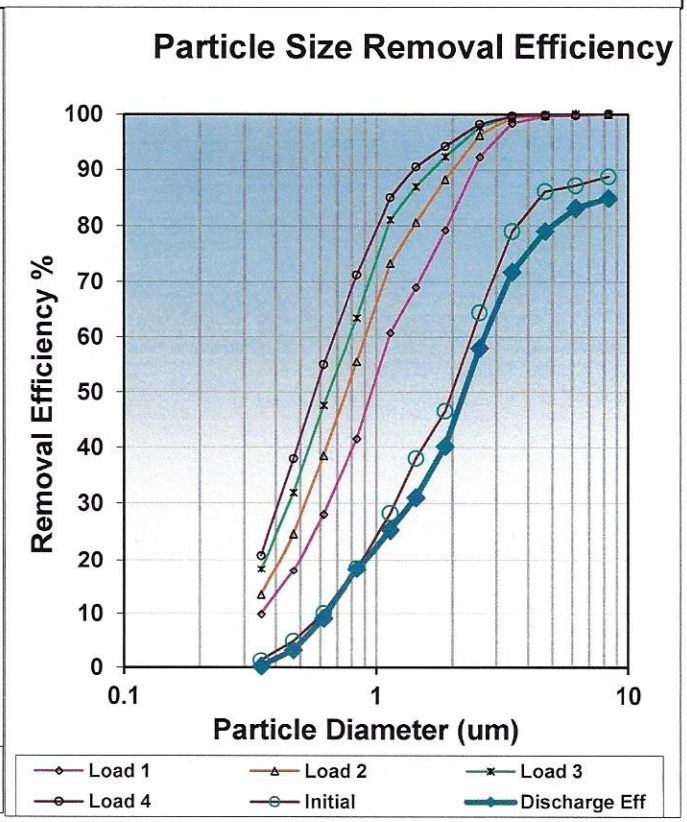
Test Performed by: JPS Approved By: Test Completed: 14-Dec-16

Blue Heaven Technologies
2820 S. English Station Rd. Louisville, KY
502 357 0132

Test No. 16-1893Rev1
Date: 14-Dec-16







Blue Heaven Technologies
2820 S. English Station Rd. Louisville, KY
502 357 0132

ASHRAE Standard 52.2-2012
Test No. 16-18 Part 1
Date: 14-Dec-16
Test Report

Data - Initial Resistance

Airflow (CFM)	Resistance (in WG)
0	0.00
492	0.04
984	0.10
1476	0.19
1968	0.28
2460	0.40

Data - Particle Removal Efficiency per Appendix J

Particle Size Range (um)	Geometric Mean Diam (um)	Particle Removal Efficiency (%)			
0.30 - 0.40	0.35	0.2	1.3	0.2	0.6
0.40 - 0.55	0.47	3.1	4.9	3.1	4.9
0.55 - 0.70	0.62	9.1	10.1	9.1	10.2
0.70 - 1.00	0.84	18.2	18.5	18.2	20.2
1.00 - 1.30	1.14	25.3	28.3	25.3	29.5
1.30 - 1.60	1.44	31.0	38.0	31.0	34.1
1.60 - 2.20	1.88	40.2	46.6	40.2	45.2
2.20 - 3.00	2.57	58.0	64.4	58.0	62.3
3.00 - 4.00	3.46	71.8	79.0	71.8	75.8
4.00 - 5.50	4.69	79.0	86.1	79.0	83.0
5.50 - 7.00	6.20	83.1	87.1	83.1	86.0
7.00 - 10.00	8.37	84.8	88.7	85.3	84.8
		Composite Min Eff (CME)	Initial	Conditioning Increment 1	Conditioning Increment 2

Conditioning Procedure Details
(included for reference as required by Appendix J11.1)

	Initial	Cond 1	Cond 2						Post
Background concentration (particles cm-3) ----->	0.78								1.01
Correlation ratio small to large particles ----->	27,677								29,389
Avg conditioning aerosol concentration (particles cm-3) ----->		58,770	57,168						
Incremental conditioning duration (minutes) ----->		1150.0	1120.0						
Cumulative conditioning duration (minutes) ----->		1150.0	2270.0						
Incremental CT (particle cm-3 min) ----->		67,585,687	64,027,684						
Cumulative conditioning CT (particles cm-3 min) ----->		67,585,687	131,613,371						
Dust Holding Capacity DHC-A (grams) ----->		NA	NA						

Data - Particle Removal Efficiency

Particle Size Range (um)	Geometric Mean Diam (um)	Particle Removal Efficiency (%)						
0.30 - 0.40	0.35	0.2	1.28	0.16	9.9	13.4	18.2	20.9
0.40 - 0.55	0.47	3.1	4.94	3.14	17.9	24.6	31.9	37.9
0.55 - 0.70	0.62	9.1	10.06	9.12	28.0	38.5	47.6	55.0
0.70 - 1.00	0.84	18.2	18.51	18.19	41.5	55.5	63.4	71.2
1.00 - 1.30	1.14	25.3	28.26	25.32	60.7	73.3	81.1	84.9
1.30 - 1.60	1.44	31.0	37.99	30.98	68.9	80.6	86.9	90.5
1.60 - 2.20	1.88	40.2	46.63	40.17	79.2	88.2	92.3	94.2
2.20 - 3.00	2.57	58.0	64.38	57.96	92.3	96.1	97.7	98.2
3.00 - 4.00	3.46	71.8	79.02	71.77	98.3	99.4	99.6	99.7
4.00 - 5.50	4.69	79.0	86.06	79.02	99.6	99.7	99.9	99.8
5.50 - 7.00	6.20	83.1	87.12	83.07	99.8	100.0	100.0	99.8
7.00 - 10.00	8.37	84.8	88.71	84.81	99.9	100.0	100.0	100.0
		CME	Initial	Discharged	Load 1	Load 2	Load 3	Load 4
Resistance after Dust Load (in w.c.) ----->				Efficiency	0.46	0.64	0.82	1.00
Dust Load (gms) ---------->					100	154	194	216

Test Data for Release Rate (%)					
Size Range No.	Geometric Mean of Particle Size Range, µm	Release Rate After Loading Stage 1	Release Rate After Loading Stage 2	Release Rate After Loading Stage 3	Release Rate After Loading Stage 4
1	0.35	0.191	0.323	0.395	0.088
2	0.47	0.057	0.072	0.098	0.028
3	0.62	0.013	0.016	0.032	0.011
4	0.84	0.009	0.012	0.012	0.010
5	1.14	0.017	0.018	0.023	0.018
6	1.44	0.010	0.012	0.013	0.041
7	1.88	0.018	0.024	0.013	0.017
8	2.57	0.016	0.034	0.014	0.021
9	3.46	0.034	0.116	0.052	0.060
10	4.69	0.035	0.109	0.050	0.082
11	6.2	0.082	0.487	0.186	0.336
12	8.37	0.000	0.310	0.266	0.442

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS
DETAILED BID SPECIFICATIONS

SECTION TWO
RING PANEL FILTERS 2-PLY

1) SCOPE

This specification covers panel filters that are a component of air-conditioning, heating, and ventilating systems.

2) PURPOSE

The purpose of this specification is to establish performance criteria and identify physical properties that are pertinent and necessary for proper filter performance. Conformance to all items in the specification is the responsibility of the bidder.

The Supplier shall agree to have a representative make on-site inspections on a quarterly basis. The purpose of this inspection is to ensure that the filters are functioning properly.

3) BID ENCLOSURES

The following items are required as part of the bid response.

3.1 A product sample for each distinctively different product design.

4) PHYSICAL CHARACTERISTICS

The filter panel shall be of a two media layer design, securely heat sealed around an internal heavy wire grid. To prevent media fluttering and sifting of trapped dust, the media layers shall have 4 internal stabilizer seals totaling not less than 60 inches in total length.

5) FILTER MEDIA

A one-half inch layer of 15 denier totaling .39 ounces per square foot shall form the upstream media. A heavily needled 6 denier media, .46 ounces per square foot, shall form the downstream media. Total media weight shall be 7.6 ounces per square yard.

The panel shall have a non-migratory, non-drying adhesive between the two layers to maximize efficiency and dust retention.

The air filter shall contain an antimicrobial biocide. Biostats are not acceptable. The antimicrobial shall effectively control the growth of mold, algae, mildew, and fungi on the filter. The antimicrobial biocide shall be part of the binder that is applied during manufacturing of the media itself.

The filter shall not support the growth of microbes. The antimicrobial must have no water solubility. The antimicrobial must be registered by the E.P.A. for the use in filters. The antimicrobial must also have clearance for use under regulations of U.S.D.A. and F.D.A. All testing on the antimicrobial must be in compliance with the American Society of Test Methods (ASTM).

The filters shall yield the following typical performance when tested by ASHRAE TEST STANDARD 52.1-1992:

TEST FILTER	24x24x1	AVERAGE EFFICIENCY	28%
RESISTANCE at 300 FPM	0.13 w.g."	AVERAGE ARRESTANCE	86%
INITIAL EFFICIENCY	12%	DUST HOLDING CAPACITY @ 1.00 w.g."	198 grams

Filter shall be MERV 7 by ASHRAE 52.2 - 1999 Independent Laboratory test at 295 fpm.

Range 1 (3.0%)	Range 2 (23.0%)	Range 3 (59.1%)
(0.3 - 1.0 microns)	(1.0 - 3.0 microns)	(3.0 - 10.0 microns)

a) **WIRE FRAME**
The internal wire support frame shall be a 9-gauge wire frame strength or greater, with internal grid support.

b) **PANEL EDGES**
Panel edges shall have media outside the wire frame grid of 1" to 1.5" of overhang to assure self-sealing (no dirt bypass).

c) **CONTROL MANUFACTURING OF END PRODUCT**
In order to control the specified panel filter product the manufacturer of the final panel product shall also be the manufacturer of the media used to make the panel filter.

6) PACKAGING AND IDENTIFICATION

Packaging shall provide adequate protection against damage or deterioration during shipment and allow complete identification of both the filters and the shipping container.

6.1 Each Filter must be marked with the following:
- 6.1.1 The name or trade name of the product/manufacturer
- 6.1.2 The biocide name on the filter

6.2 Shipping containers shall be marked with the following:
- 6.2.1 The name or trade name of the product
- 6.2.2 The filter sizes
- 6.2.3 The carton quantity
- 6.2.4 The manufacturer part number
- 6.2.5 Underwriters' Laboratories, Inc. official listing mark including notation of class.

FIBERBOND

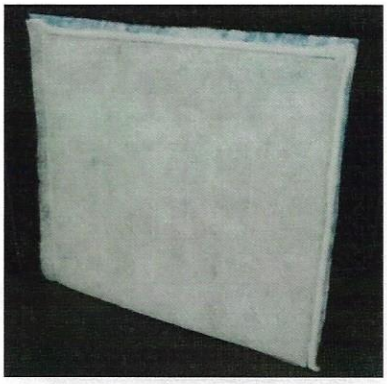
RESEARCH AND DEVELOPMENT LABORATORY

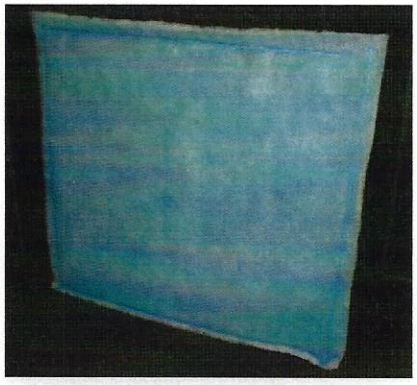
110 Menke Road Michigan City, IN 46360
Tel: (219) 879-4541 Fax: (219) 874-7502

ASHRAE Standard 52.2-2012 (M) - Addenda 2015 Test Report
Initial Efficiency & Resistance - Dust Holding Capacity

Test Date: 10/24/13 **Test #** 334

Filter Data		Test Conditions	
Manufacturer:	Fiber Bond Corporation	**Temperature:**	68°
Filter:	Poly Panel	**Humidity:**	35%
Part Number:	250002424	**Test Aerosol:**	KCl, Neutralized
Size:	24 x 24 x 1		
Media Area:	4 ft²		
Adhesive:	Yes	**Test Operator:**	J. Mazur
Antimicrobial:	None		





Test Results

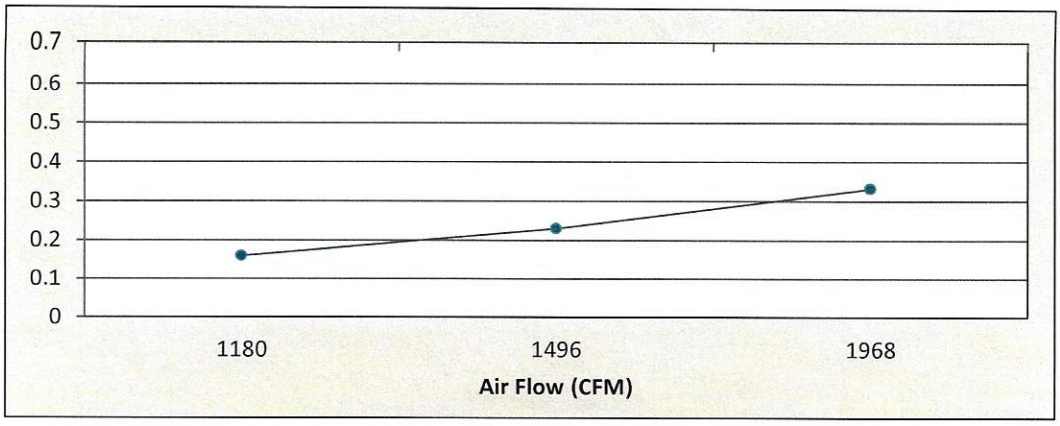
Airflow Rate (CFM):	1,180
Air Velocity (FPM):	295
Initial Resistance:	0.16" w.g.
Final Resistance:	1.0" w.g.
Initial Efficiency (0.3 - 1.0 microns):	11%
Initial Efficiency (1.0 - 3.0 microns):	29%
Initial Efficiency (3.0 - 10.0 microns):	54%
Dust Holding Capacity:	222 grams
Average Arrestance:	87%
Minimum Efficiency Reporting Value (MERV):	**7**

**If Initial Data is Minimum*

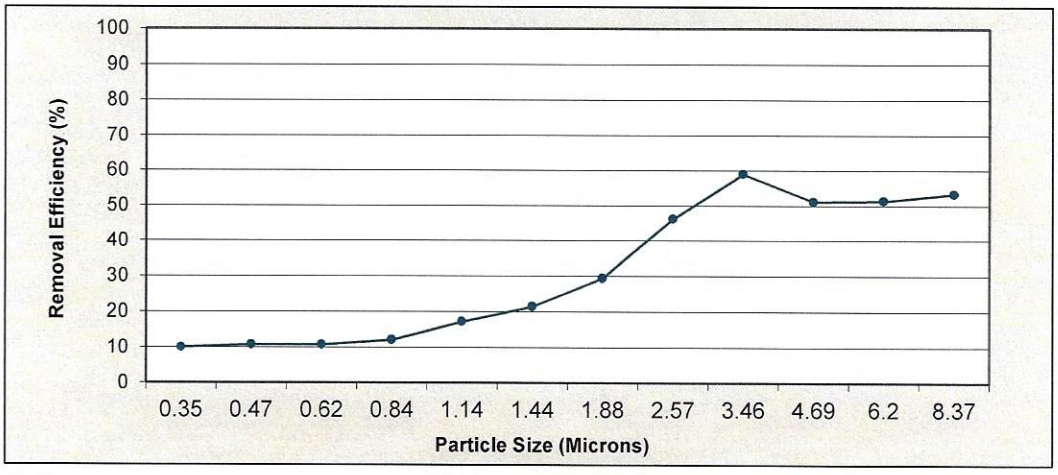
ASHRAE STANDARD 52.2(M) TEST REPORT
TECHNICAL DATA

Test Date: 10/24/13 **Test # 334**

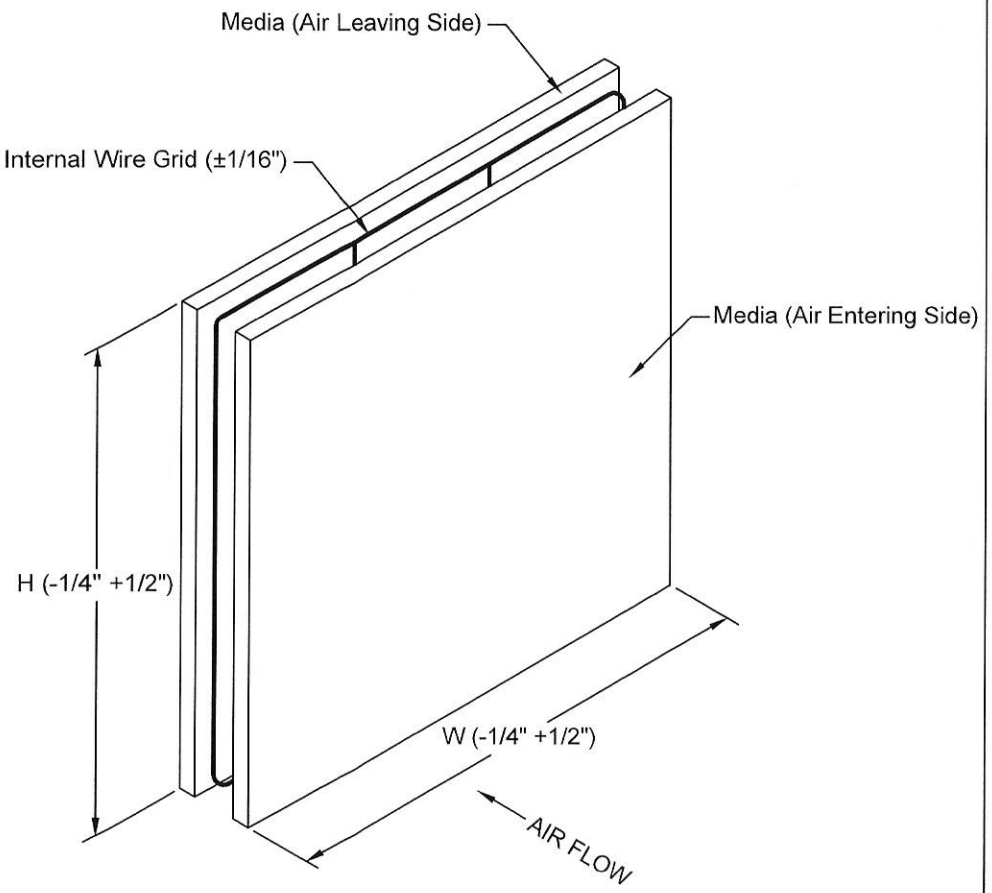
RESISTANCE VS. AIR FLOW
CLEAN DEVICE

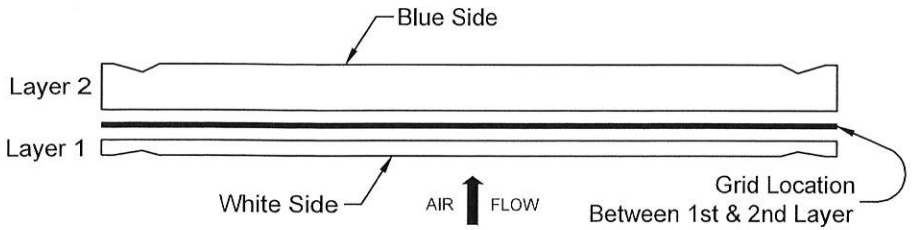


INITIAL PARTICLE REMOVAL EFFICIENCY



Particle Size Range (microns)	Mean Particle Size	Initial Removal Efficiency (%)
.30 - 0.40	0.35	10
.40 - 0.55	0.47	10.7
.55 - .70	0.62	10.7
.70 - 1.0	0.84	12.1
1.0 - 1.3	1.14	17.2
1.3 - 1.6	1.44	21.4
1.6 - 2.2	1.88	29.3
2.2 - 3.0	2.57	46.3
3.0 - 4.0	3.46	59
4.0 - 5.5	4.69	51.2
5.5 - 7.0	6.2	51.4
7.0 - 10.0	8.37	53.3





1. General

1.1. Medium efficiency panel filter shall be Poly Panel as manufactured by Fiber Bond Corporation.

1.2. Sizes shall be noted on supporting materials.

2. Construction

2.1. Panel filter media shall be polyester fiber bonded with a binder to provide a media that exhibits high strength and durability.

2.2. The panel filter shall consist of two distinct media layers for maximum depth loading to optimize service life.

2.3. The air entering side of the panel is white. The air exiting side of the panel is blue to ensure correct installation.

2.4. The panel filter shall contain an internal wire support grid that is 3/4" under nominal panel size and sealed within the media.

2.5. The perimeter of the internal wire grid shall be constructed of 9 gauge wire. Where applicable, 11 gauge cross wires shall be used for additional support.

2.6. To prevent bowing and fluttering of the filter while in use, most filter sizes shall have four internal seals of at least 60 inches in combined length. When the filter has a dimension of 16 inches or less the four internal seals shall have a combined length of at least 30 inches.

2.7. The panel filter shall have a nominal thickness of 1".

3. Performance

3.1. The panel filter shall contain non-drying tackifier on the air leaving side which locks in dust particles.

3.2. The panel filter shall be MERV 7 in accordance with ASHRAE 52.2-2012 M with 2015 Addenda, when tested at 295 fpm.

3.3. The panel filter shall have a typical initial resistance of 0.16" w.g. at 295 fpm and a recommended final resistance of 1.0" w.g.

3.4. The panel filter shall withstand operating temperatures up to 200° F.

3.5. The panel filter shall be UL900 Classified by Underwriters Laboratories as to flammability.

FIBER BOND
110 MENKE ROAD
MICHIGAN CITY, IN 46360
P: 219-879-4541
F: 219-874-7502

SIZE	DWG NO.	REV	SHEET
A	P101	7	1 OF 1
SCALE N/A	BY B. PAGE	DATE 3/8/16	

TECHNICAL SPECIFICATIONS

POLY PANEL

FIBERBOND

The **Best** Filters Begin With The **Best** Media

Polyester Media

Polyester Media Technical Data

Polyester Media

FM Series:
Dry Media .5", 1" & 2"
Average Arrestance
82%

FP Series:
Tackified Media .5", 1" & 2"
Average Arrestance
85%

Maximum Operating Temperature:
200° F

Flammability:
UL 900 Classified

Media	Characteristics	Color	Nominal Thickness	Roll Length	Resistance 295 fpm	MERV Value
FM 25	Dry Media	White	.25"	270'	0.04" w.g.	5
FM 50	Dry Media	White	.50"	135'	0.09" w.g.	5
FM 100	Dry Media	White	1"	90'	0.13" w.g.	6
Poly II	Dry Media	Yellow & White	2"	60'	0.13" w.g.	6
FP 50	Tackified Media	Blue & White	.50"	135'	0.10" w.g.	6
FP 100	Tackified Media	Blue & White	1"	90'	0.12" w.g.	7
FP 200	Tackified Media	Blue & White	2"	60'	0.14" w.g.	7

Media test results in accordance with internal Fiber Bond test method ASHRAE 52.2 - 2012 (M) November 2013

Fiber Bond has a policy of continuous product research and improvement and reserves the right to alter design and specifications without notice.

FIBERBOND

110 Menke Road, Michigan City, IN 46360 • Phone (219) 879-4541 • Fax (219) 874-7502
Email: customer.service@fiberbond.net • www.fiberbond.net

FIBERBOND

The **Best** Filters Begin With The **Best** Media

Polyester Media

Fiber Bond Polyester Media

FM Dry Media • FP Tackified Media • Poly II Dry Media

Polyester Media

Low Initial Resistance

- .25", .5", 1" & 2" Nominal Thicknesses
- Bulk Media Rolls Up To 92" Wide
- Pre-Cut Pads
- Poly-Perf®
- Fiber blend includes regenerated fibers

FM Series

Dry Polyester Media

FM 25, FM 50, FM 100 & Poly II

FP Series

Tackified Polyester Media

FP 50, FP 100 & FP 200

Consistently Top-Of-The-Line

Polyester Media

Polyester Media Manufactured On-Site

Fiber Bond manufactures all of its own filtration media, assuring that it is closely monitored for quality and consistency. Because everything is done under one roof, Fiber Bond's lead times are among the industry's best – shipping customer orders in far less time than other manufacturers.

Top-Quality, Affordable And Functional

Fiber Bond polyester media is produced using an air laid process that creates uniform, high-loft media. The manufacturing process delivers a media that offers true depth-loading. The highly resilient fibers lend themselves to vacuum-packing which significantly reduces landed cost.

Economical, Dependable Performance

Polyester media is ideal for applications that need dependable filtration and economy. Polyester media is a great choice for pad and frame, or blanket usage.

FIBERBOND

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Email: customer.service@fiberbond.net • www.fiberbond.net

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS
DETAILED BID SPECIFICATIONS

SECTION FOUR
HIGH-EFFICIENCY ASHRAE GRADE POCKET FILTER

1.0 GENERAL

1.1 Air filters shall be high efficiency, extended surface pocket style filters consisting of high loft air laid micro fine glass media, a reinforced ABS plastic header, ABS plastic pocket retainers, and bonding agents to prevent air bypass and ensure leak free performance.

1.2 Sizes shall be as noted on drawings or other supporting materials.

2.0 CONSTRUCTION

2.1 Filter media shall consist of high-density air laid lofted micro fine glass media that is chemically bonded to a synthetic micro mesh media support backing forming a lofted filter blanket.

2.2 Individual pockets shall contain a minimum of 40 stitching support points per square foot of media area. The sides and ends of each pocket shall be sewn with a chain-link over lock stitch.

2.3 Pockets shall be formed into tapered pleats, supported by controlled media space stitching, to promote uniform airflow across the surface of the media. At any point, the sizes of the upstream and downstream passages shall be proportional to the volume of filtered air. The pockets shall also have a conical configuration to minimize contact with HVAC system components.

2.4 Support members shall include an ABS plastic header and ABS plastic pocket retainers. The header shall be joined to the media to prevent air bypass. Individual pocket retainers shall be attached to the header frame with anchor ports allowing for visual confirmation. Bypass between pockets shall be eliminated through a snap-to-seal pocket retainer that shall be an integral part of the two-piece header design. The frame shall form a rigid and durable support assembly.

2.5 The air exiting side of the air tunnels include a pocket flange to ensure pocket integrity throughout the life of the filter. A downstream pocket-to-pocket partition shall provide additional pocket separation to ensure full flow through the entire media area.

2.6 A filter-to-filter sealing gasket shall be installed on one of the vertical members of the filter header.

3.0 PERFORMANCE

3.1 The filter shall have a Minimum Efficiency Reporting Value of MERV (11, 13, 14, 16) when evaluated under the guidelines of ASHRAE Standard 52.2-2012. It shall also have a MERV-A rating of (11, 13, 14, 15) when evaluated under ASHRAE Standard 52.2, Appendix J. It shall have an efficiency of (ePM10-70, ePM1-60, ePM1-70, ePM1-80) when evaluated per ISO filter testing standard 16890.

3.2 Initial resistance to airflow as listed by the manufacturer on a 22" depth 10-pocket bag shall be (0.24", 0.38", 0.45", 0.62")* w.g at an airflow of 500 fpm. Additional information shall be as noted on drawings or other supporting materials.

3.3 The manufacturer shall warranty that the filter shall be capable of withstanding 10.0" w.g. without failure of the filter.

3.4 The filter shall be classified by Underwriters Laboratories as UL 900.

3.5 Manufacturer shall provide evidence of facility certification to ISO 9001:2015.

Supporting Data – Provide product test reports for each listed efficiency including all details as prescribed in ASHRAE Standards 52.2 and ISO Standard 16890

SUPPORTING DATA – Provide ASHRAE product test report per ASHRAE Standard 52.2, including testing per appendix J.

4.0 HIGH EFFICIENCY BAG FILTER SUPPORTING DATA – Provide ASHRAE product test report per ASHRAE Standard 52.2 including testing per appendix J.

4.1 NOMINAL FILTER SIZE (WxHxD) 24x24x22	
Nominal Average Dust Spot Efficiency (%)	80-85%
Gross Media Area	71
4.2 TEST CONDITIONS	
Filter Face Velocity (FPM)	493
Test Air Flow Rate (CFM)	1968
Maximum Initial Resistance (Inches W.G.)	0.37"
Final Resistance (Inches W.G.)	1.5
Dust Holding Capacity under 52.2-2012	444 grams
4.3 PERFORMANCE REQUIREMENTS	
Minimum Composite Average Efficiencies (%) ASHRAE STANDARD 52.2-2012 with Appendix J	E1: 71-A; E2: 97-A E3: 100-A
Minimum Efficiency Reporting Value (MERV-A)	13/A

5.0 BID ENCLOSURES

The following items are required as part of the bid response.

5.1 For the master final filter size of 24X24X22, there must be one (1) independent laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012 MERV 13/13A. Permission shall be granted by the bidder to verbally verify the performance results with the independent laboratory that performed the tests.

5.2 A product sample for each distinctively different product design.

Camfil Farr
1 North Corporate Drive
Riverdale, NJ 07457
(973) 616-7300

ASHRAE® 52.2-2012 Test Results

Filter Description

Manufacturer	Filter Model	Filter Part Number
Camfil Farr	HF-ES M13/24/24/22/10	405618A22

Filter Type	Media Type	Media Color	Media Area (ft^2)
Pocket Filter	Fiberglass/Air Laid	Green	71

Nominal Size	Actual Size						
HxWxD	Hgt (in)	Wid (in)	Dep (in)	Pleats/Pockets	UL Rating	LabFile	CrRef
24x24x22	23 3/8	23 3/8	22	10	Class 2	1414	1414

ASHRAE® 52.2-2012

Date:	05/09/2013
Test Number:	2013-0509-1334

Test Data	Test Results	Rated Values	
AirFlow (cfm)	1970	1970	
Nominal Vel (fpm)	493	493	
Initial ΔP (inWG)	0.37	0.40	
MERV	13	13	@1970cfm
E_1 (0.3-1.0μm), (%)	70	n/a	
E_2 (1.0-3.0μm), (%)	97	≥ 90	
E_3 (3.0-10μm), (%)	100	≥ 90	
Final DP, (inWG)	1.5	1.5	
Arrestance, (%)	100	97	
DHC, (g)	339		

Temp, (F)	80	Test Aerosol	KCL
RH, (%)	40	Loading Dust	ASHRAE

Comments

Certified Test Per ASHRAE 52.2-2007 (B)

Approval: Don Thornburg
Electronic Signature, Original on file

Operator	Particle Counter	Model
A. Acain	Climet	CI-500

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Riverdale, NJ 07457
(973) 616-7300

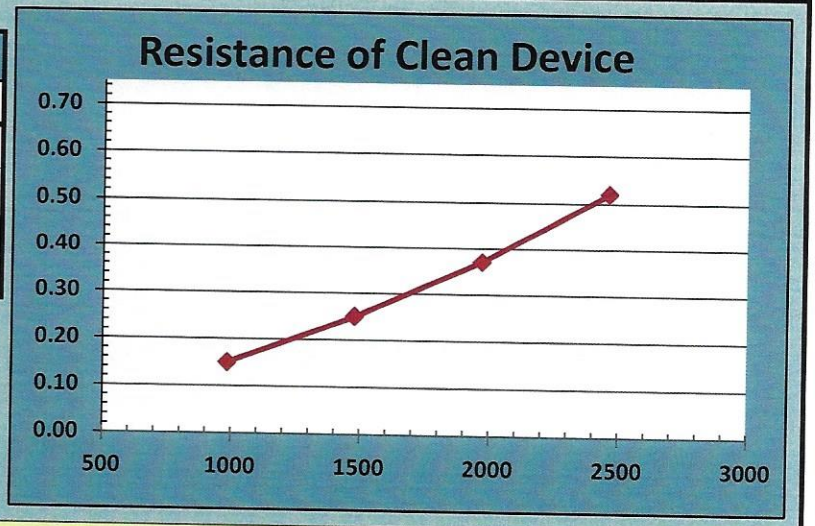
ASHRAE® 52.2-2012 Test Results

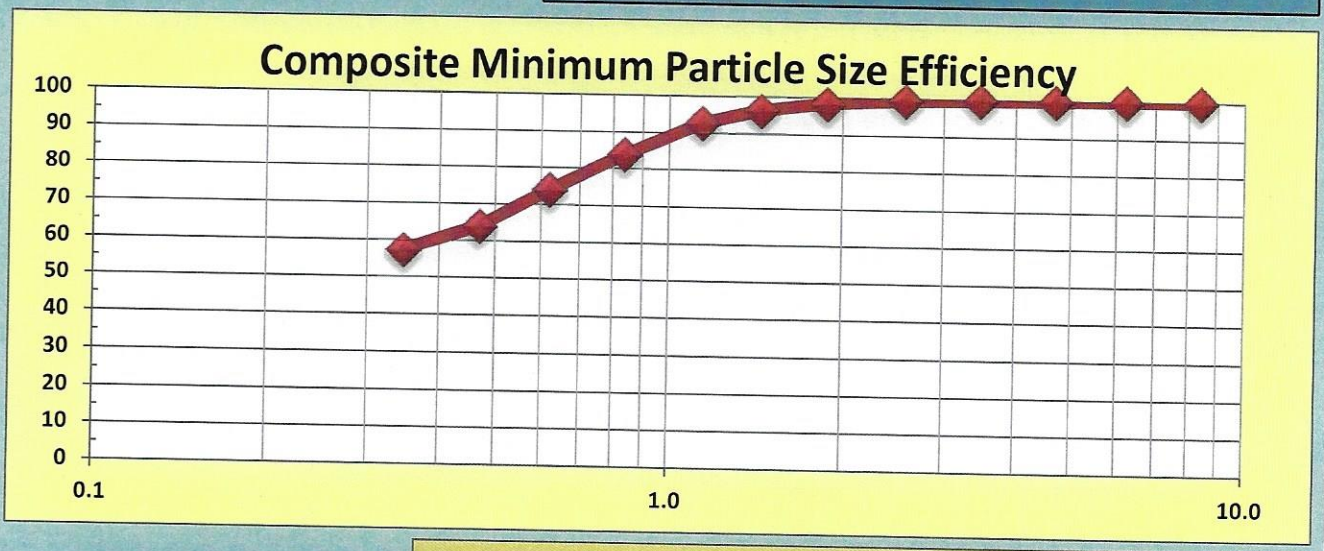
Date:	05/09/2013	Test Number:	2013-0509-1334

Manufacturer	Filter Model	Filter Part Number
Camfil Farr	HF-ES M13/24/24/22/10	405618A22

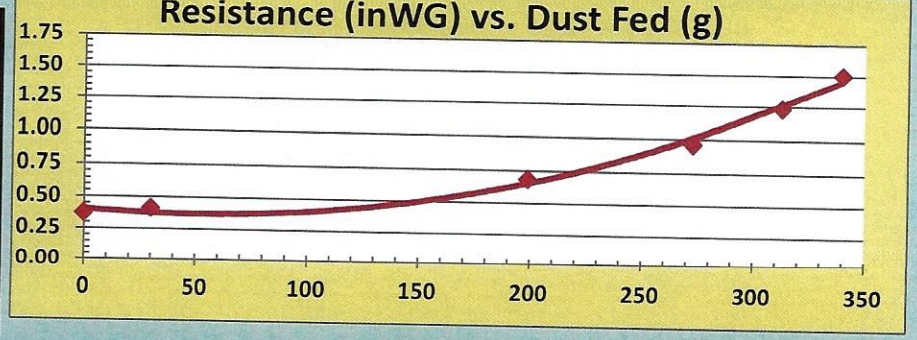
Clean Resistance Table

cfm	m^3/s	inWG	Pa
985	0.46	0.15	37
1478	0.70	0.25	62
1970	0.93	0.37	92
2462	1.16	0.52	129





ΔP (inWG)	Dust Fed (g)
0.37	0
0.41	30
0.68	199
0.96	273
1.23	313
1.50	340



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1 North Corporate Drive
Riverdale, NJ 07457
(973) 616-7300

ASHRAE® 52.2-2012 Test Results

Date	05/09/2013	Test Number	2013-0509-1334
Manufacturer		Filter Model	Filter Part Number
Camfil Farr		HF-ES M13/24/24/22/10	405618A22

Efficiency Data Table	1	2	3	4	5	6	7	8	9	10	11	12	
Minimum Dia (µm)	0.3	0.4	0.55	0.7	1.0	1.3	1.6	2.2	3.0	4.0	5.5	7.0	0.35
Maximum Dia (µm)	0.4	0.55	0.7	1.0	1.3	1.6	2.2	3.0	4.0	5.5	7.0	10.0	0.45
Geometric Mean (µm)	0.35	0.47	0.62	0.84	1.14	1.44	1.88	2.57	3.46	4.69	6.20	8.37	0.40
Initial PSE	57	64	74	84	93	96	98	100	100	100	100	100	60
Dust Cond PSE	66	72	81	90	96	98	99	100	100	100	100	100	68
1st Dust Load PSE	88	92	95	98	99	100	100	100	100	100	100	100	90
2nd Dust Load PSE	93	96	98	99	100	100	100	100	100	100	100	100	94
3rd Dust Load PSE	95	97	99	99	100	100	100	100	100	100	100	100	96
4th Dust Load PSE	96	98	99	100	100	100	100	100	100	100	100	100	97
Composite Min PSE	57	64	74	84	93	96	98	100	100	100	100	100	60

All testing is performed in compliance with the latest revision of ASHRAE Standard 52.2 in a state-of-the-art laboratory facility located at Camfil Farr's U.S. corporate headquarters in Riverdale, NJ. Test Reports without DHC values show efficiency only and the associated MERV value is a estimated value based on the single efficiency data shown. For more information, contact your local Camfil Farr representative or distributor.

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Blue Heaven Technologies

2820 S. English Station Road - Louisville, KY 40299

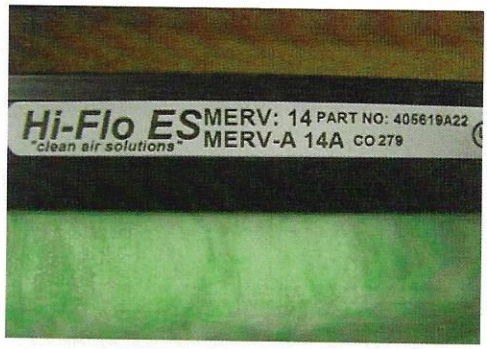
Tel: (502) 357-0132 Fax (502) 267-8379

Date: 22-May-12 TEST NO. 12-0631

ASHRAE Standard 52.2-2007
APPENDIX J TEST REPORT
Efficiency with KCL Conditioning

Filter Description

Manufacturer	Camfil Farr
Filter Model	HFES M14
Part Number	405619A22
Generic Filter Type	Bag Filter
Nominal Dimensions (H x W x D)	24"x24"x22"
Pocket / Pleat Quantity	10 Pockets
Media Type	Air Laid Glass
Est. Gross Media Area	71.45 Sq/Ft
Adhesive Type	N/A



Test Conditions

Conditioning Aerosol Type	KCL		
Loading Dust Type	N A	Test Air Temp (degrees F.)	70
Barometric Pressure (In. Hg.)	29.44	Relative Humidity (%)	50

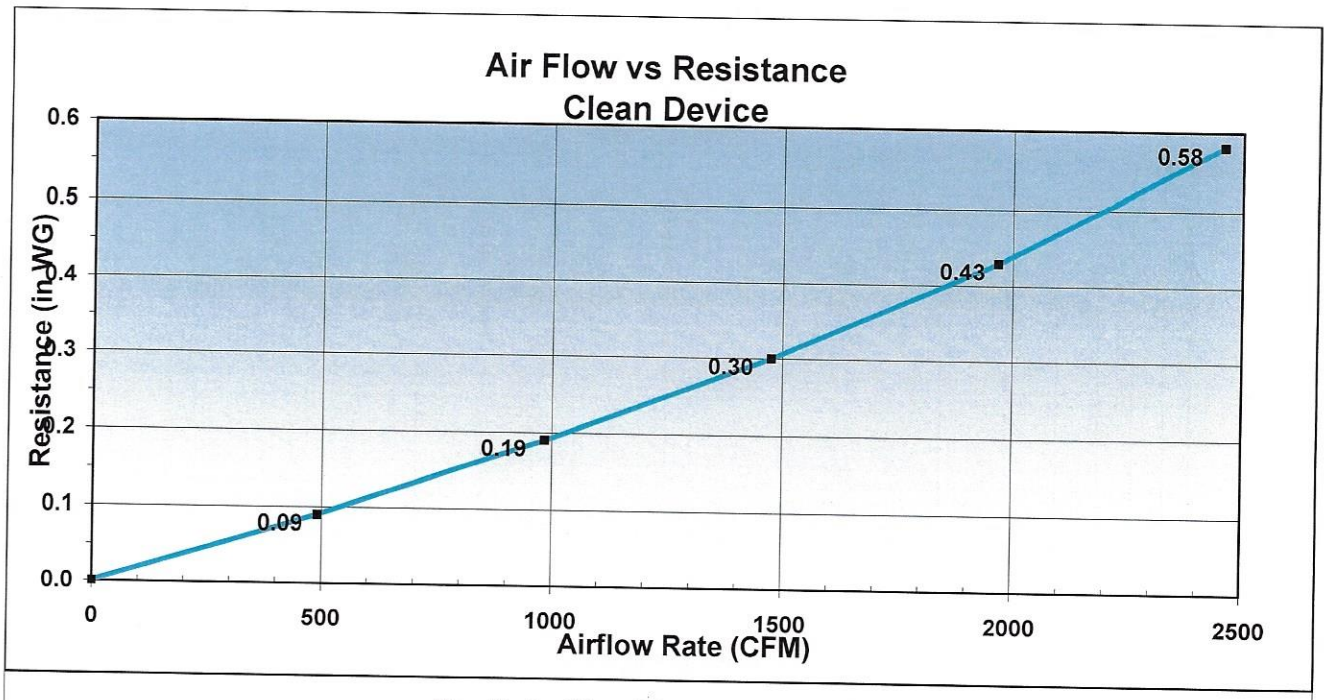
Test Results

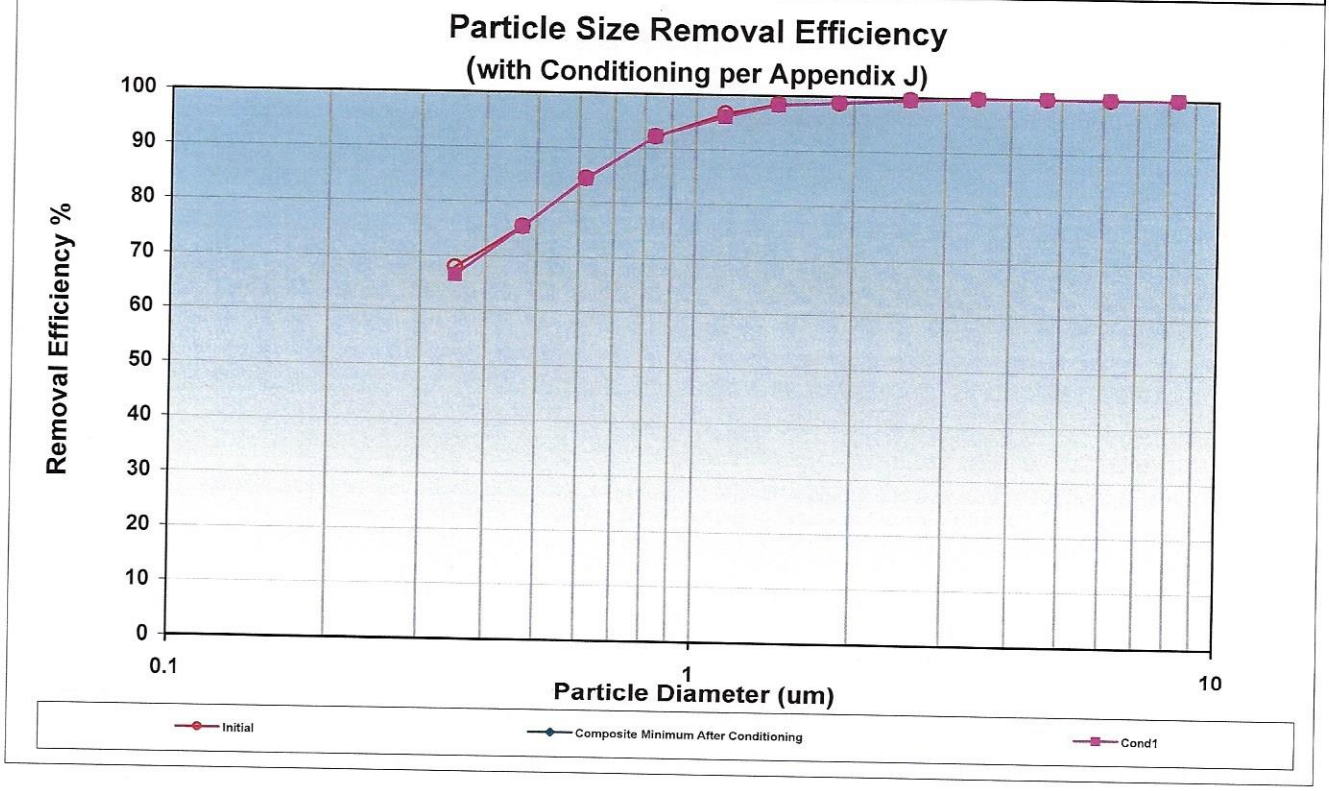
Airflow Rate (CFM)	**1968**
Nominal Face Velocity (fpm)	**492**
Initial Resistance (in WG)	**0.43**
E1-A (%) Min. Composite Efficiency after KCL Conditioning 0.30 - 1.0 um	**80**
E2-A (%) Min. Composite Efficiency after KCL Conditioning 1.0 - 3.0 um	**98**
E3-A (%) Min. Composite Efficiency after KCL Conditioning 3.0 - 10.0 um	**100**
***Minimum Efficiency Reporting Value after KCL Conditioning (MERV-A)** (* assumes no efficiency reduction during subsequent ASHRAE dust loading)	**MERV 14-A @ 1968 CFM**

Comments Tested For: Camfil Farr

Approval: R.J.B. Bell

Test No. 12-0631
Date: 22-May-12





Test No. 12-0631
Date: 22-May-12

Data - Initial Resistance

Airflow (CFM)	Resistance (in WG)
0	0.00
492	0.09
984	0.19
1476	0.30
1968	0.43
2460	0.58

Data - Particle Removal Efficiency per Appendix J

Particle Size Range (um)	Geometric Mean Diam (um)	Particle Removal Efficiency (%)		
0.30 - 0.40	0.35	66.65	68.00	66.65
0.40 - 0.55	0.47	75.53	75.57	75.53
0.55 - 0.70	0.62	84.36	84.45	84.36
0.70 - 1.00	0.84	92.24	92.25	92.24
1.00 - 1.30	1.14	96.05	96.59	96.05
1.30 - 1.60	1.44	98.23	98.38	98.23
1.60 - 2.20	1.88	98.72	98.72	98.83
2.20 - 3.00	2.57	99.42	99.58	99.42
3.00 - 4.00	3.46	99.90	99.91	99.90
4.00 - 5.50	4.69	99.94	99.94	100.00
5.50 - 7.00	6.20	99.92	99.92	100.00
7.00 - 10.00	8.37	100.00	100.00	100.00
		Composite Min Eff (CME)	Initial	Conditioning Increment 1

Conditioning Procedure Details

(included for reference as required by Appendix J11.1)

	Initial	Cond 1	Post	Req'd by App J
Background concentration (particles cm-3) ----->	0.27		1.40	low
Correlation ratio small to large particles ----->	62,947		28,361	> 20,000
Avg conditioning aerosol concentration (particles cm-3) ----->		209,096		< 1,000,000
Incremental conditioning duration (minutes) ----->		435.0		record
Cumulative conditioning duration (minutes) ----->		435.0		
Incremental CT (particle cm-3 min) ----->		122,530,500		> 6.4 x 10E7
Cumulative conditioning CT (particles cm-3 min) ----->		122,530,500		< 1.2 x 10E9
Dust Holding Capacity DHC-A (grams) ----->		NA		record

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS
DETAILED BID SPECIFICATIONS

SECTION THREE
HVAC MINI-PLEAT 4V-CELL TYPE
FILTER U.L. CLASS 2

1.0 GENERAL

1.1 Air filters shall be high-efficiency ASHRAE pleat-in-pleat V-bank disposable type assembled in a compact and secure enclosing frame.

1.2 Sizes shall be as noted on drawings or other supporting materials.

2.0 CONSTRUCTION

2.1 Filter media shall be of micro fine glass formed into uniformly spaced pleats separated by hot melt separators and formed into a mini-pleat pack design.

2.2 Each mini-pleat pack shall be assembled into a V-bank configuration with an appropriate number of packs to obtain required pressure drop.

2.3 The media packs shall be bonded to the inside periphery of the enclosing frame with a fire-retardant phosphorus-free sealant.

2.4 The enclosing frame should be comprised of ABS plastic.

2.5 Rigid plastic end caps shall be mechanically fastened to the top and bottom of the enclosing structure to ensure a rigid and durable filter.

2.6 Front face construction shall consist of a ¼" to 1" center point space between the media pack and front face of filter frame, thus minimizing air turbulence and pressure drop growth between pleat pre-filter and final filter.

3.0 PERFORMANCE

3.1 The filter shall have a Minimum Efficiency Reporting Value of MERV 13 when evaluated under the guidelines of ASHRAE Standard 52.2. It shall have a MERV-A of 13 when tested using appendix J of that standard.

3.2 Initial resistance to airflow shall be 0.27".64" w.g at an airflow of 500 fpm.

3.3 Filter shall be rated by Underwriters Laboratories as UL Class 900.

3.4 Manufacturer shall provide evidence of facility certification to ISO 9001:2008.

SUPPORTING DATA - Provide ASHRAE product test report per ASHRAE Standard 52.2 including testing per appendix J.

3.4.1	NOMINAL FILTER SIZE (WxHxD)	24x24x12
	Nominal Average Dust Spot Efficiency (%)	90-95%
	Gross Media Area	200
3.4.2	TEST CONDITIONS	
	Filter Face Velocity (FPM)	492
	Test Air Flow Rate (CFM)	1968
	Maximum Initial Resistance (Inches W.G.)	0.27"
	Final Resistance (Inches W.G.)	1.5
	Dust Holding Capacity under 52.2-2012	350 grams
3.4.3	PERFORMANCE REQUIREMENTS	
	Minimum Composite Average Efficiencies (%)	E1: 69; E2: 92; E3: 99
	ASHRAE STANDARD 52.2-2012 with Appendix J	
	Minimum Efficiency Reporting Value (MERV-A)	13/A

BID ENCLOSURES

The following items are required as part of the bid response.

4.1 For the master final filter size of 24X24X12, there must be one (1) independent laboratory test report showing performance results when tested to ASHRAE standard 52.2-2012 MERV 13/13A. Permission shall be granted by the bidder to verbally verify the performance results with the independent laboratory that performed the tests.

4.2 A product sample for each distinctively different product design.

Hi-Flo® ES

Energy Saving, Extended Surface Area, High Efficiency Air Filter

Air filters are the first line of defense to protect people and processes in buildings. The Camfil Hi-Flo ES can remove contaminants including fumes, smoke, bacteria, fungi, and virus-bearing droplet nuclei. The Hi-Flo ES is also the filter of choice for the removal of nuisance contaminants such as pollen, paper dust, and other atmospheric impurities.

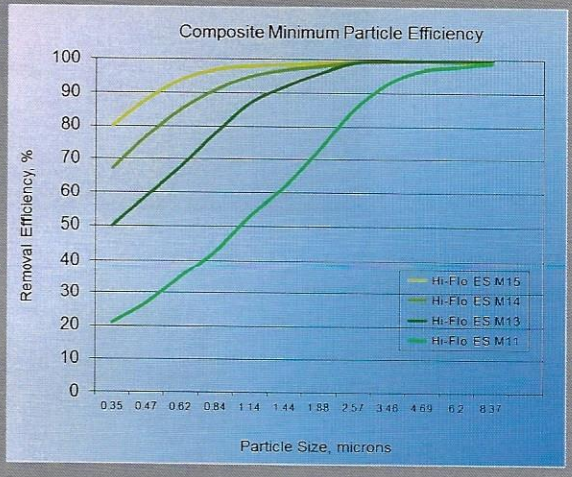
Hi-Flo ES filters are available in four efficiencies: MERV 11, MERV 13, MERV 14 and MERV 15, when evaluated per ASHRAE Standard 52.2. The Hi-Flo ES also has a MERV-A value of 11A, 13A, 14A and 15A, respectively when tested per Appendix J of the same Standard, ensuring that the Hi-Flo ES will provide maintained particle capture efficiency throughout the life of the filter.

Air filters are the most significant component of an HVAC system that should be considered for total cost of ownership. The Hi-Flo ES:

- Has the lowest operating cost in terms of energy usage. Energy cost per filter can be as high as three times the cost of the filter itself. The Hi-Flo ES air filter's low maintained pressure drop can save over 30% of electric utility costs when compared to other filters.
- Requires less filter changes than other high efficiency filters. Savings include lower labor costs to change filters, decreased disposal costs, less space in landfills, and a lower carbon footprint.

The Camfil Hi-Flo ES 5-Star ECI rating ensures maintained efficiency and a longer service life than same class high efficiency filters. Its sustainable features meet the green demands of building owners at the lowest cost of ownership. Performance is also guaranteed!

Energy saving pocket filter with guaranteed lifetime efficiency.

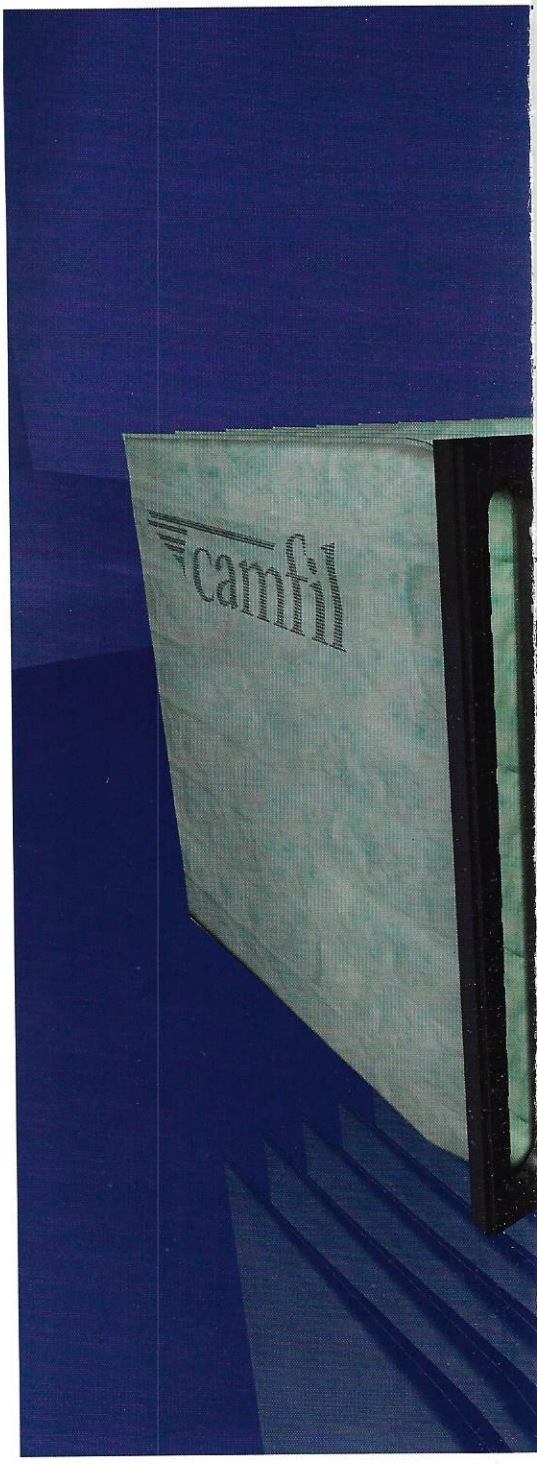


The above chart shows relative efficiency values at various particle sizes when tested in accordance with ASHRAE Standard 52.2-2007. When tested in accordance with Appendix J of that Standard, the Hi-Flo ES maintains these efficiency values throughout the life of the filter.

[1] A 5-Star rating indicates that this filter performs in the top 20% of all products of similar construction in the HVAC industry. Factors of consideration include maintained efficiency, energy usage and resistance to air flow. Detailed evaluation information is available from your Camfil sales outlet or on the web at www.camfil.com.

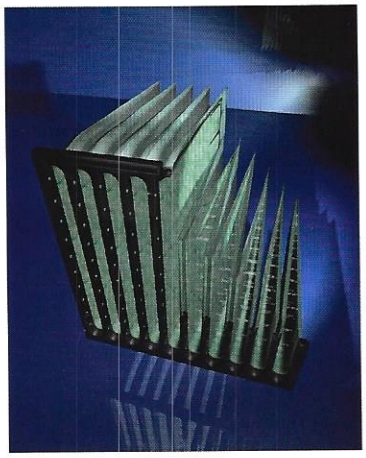
The Hi-Flo ES incorporates exclusive Camfil air laid microfiber glass media that ensures reliable efficiency throughout the life of the filter. Its fine fiber diameter and uniform loft results in a consistent sub-micron particle capture and a low resistance to airflow. This exclusive media is designed to maintain this low resistance to air flow, saving energy, while still holding efficiency throughout the filter life. The Hi-Flo ES will maintain its particle efficiency, regardless of dust loading and/or humidity.

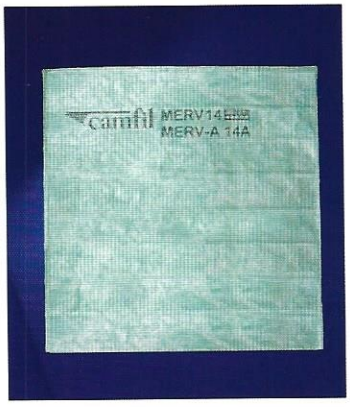
A synthetic micro mesh media backing ensures media protection and support in turbulent or varying airflows.



Camfil is the only manufacturer to use tapered pocket stitching — pockets are stitched to prevent pocket contact throughout the entire depth of the filter, ensuring uniform airflow and allowing full use of the media area. This results in a longer filter life, lower HVAC energy costs, less filter changes, lower labor costs, lower disposal costs and an overall greener, and environmentally-friendly product.

Pocket stitching is sealed to eliminate air bypass through stitching points. This unique sealant maintains a flexibility that is unaffected by turbulence or varying airflows.

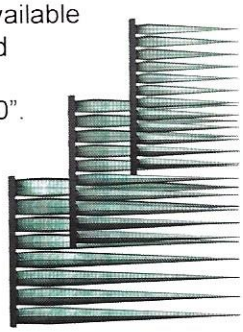




The Hi-Flo ES pockets are also tapered from the air entering side of the filter to the air exiting side of the filter. This conical pocket configuration also prevents media contact against duct interiors.

Each filter is identified on the filter as to its MERV and MERV-A.

The Hi-Flo ES is available in 4 efficiencies and 3 pocket depths, 12", 15", 22" and 30".



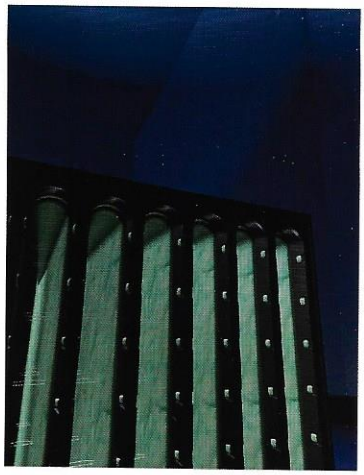
The Camfil Hi-Flo ES (Energy Saver) comes fully guaranteed [1] to outperform all competitive products of its kind and to deliver the highest energy savings possible in the industry while maintaining its rated efficiency.

This guarantee eliminates associated risks with choosing or converting to the Hi-Flo ES and serves as proof that Camfil stands behind the product's design features and performance capabilities.

[1] Includes all models with 20" depth or longer.

The reinforced ABS plastic header frame is assembled from matching halves to provide rigid and durable filter support. Frame racking is eliminated and the filter fits securely into the side-access housing or built-up bank holding frame. Its rigidity reduces the possibility of air bypass, even during turbulent airflow. One vertical header includes a gasket to prevent air bypass between filters when they are installed in a filter track.

Each air tunnel on the air entering side of each pocket is formed to promote uniform airflow through the entire length of the pocket.



Filter bypass between pockets is eliminated through a unique snap-to-seal pocket retainer feature that is an integral part of the 2-piece header design. The media pocket is securely attached to the header frame with anchor ports allowing for visual confirmation.

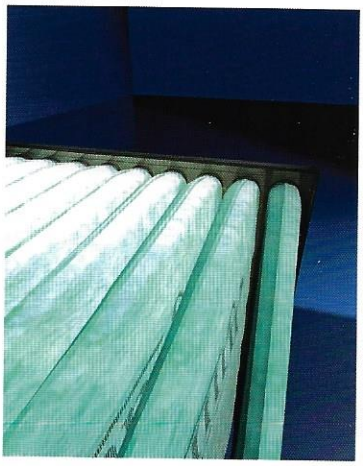
The snap-together design of the header results in frame junctions that are completely enclosed.

Sharp corners (see inset) are eliminated for the protection of service personnel. Pocket damage, or other damage related to sharp metal edges or projections is prevented.

Exclusive pocket guard protects pockets during installation — pockets are isolated and not subject to damage or tearing during installation.

The air exiting side of the air tunnels include a pocket flange to ensure pocket integrity throughout the life of the filter. Pockets are also protected during turbulent air flow.

A downstream pocket-to-pocket partition provides additional pocket separation to ensure full flow through the entire media area.

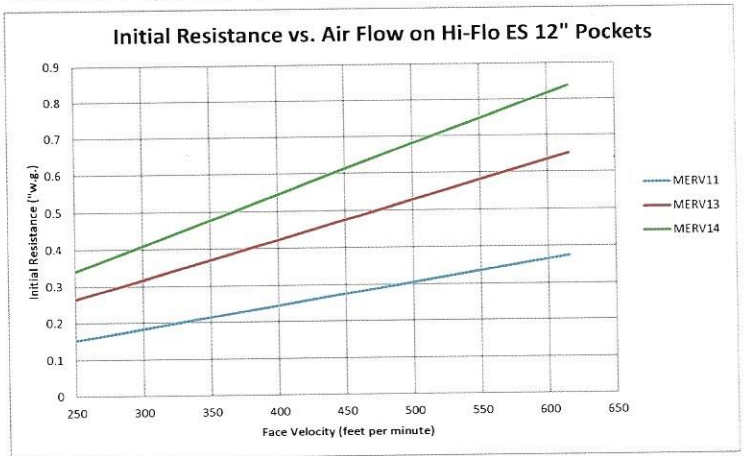


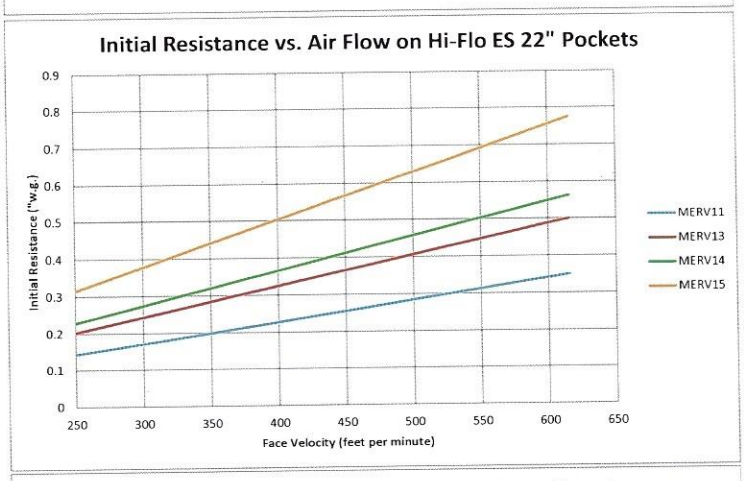
Hi-Flo® ES

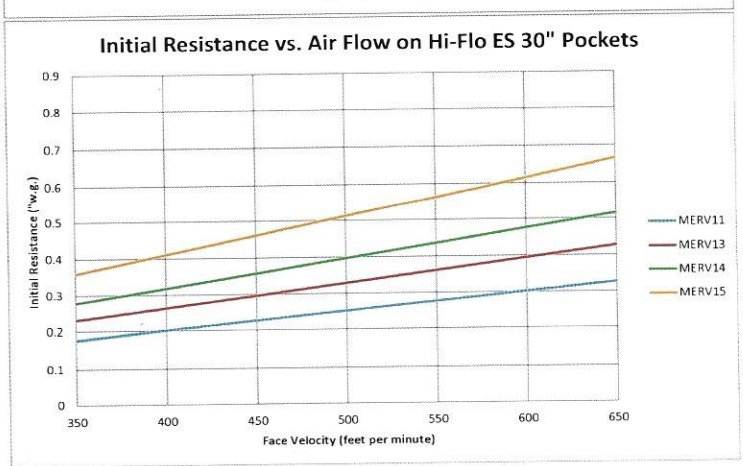
Energy Saving, Extended Surface Area, High Efficiency Air Filter

Initial Resistance Versus Airflow

Contact factory before operating outside of airflow region.



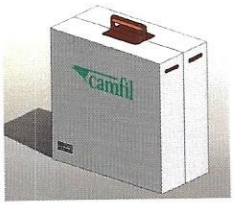




Contact factory for information on 15" deep model.

Options:

Standard Hi-Flo ES includes filter-to-filter sealing gasket on one vertical header side. Gasket or additional gaskets are available on all header sides and on face of filter. Contact factory.



All Hi-Flo ES filters are shipped in an easy-to-handle container that includes a transport handle. Filter service personnel can easily transport eight filters and discard the old filters in the same container. Filter change is easy and disposal costs are reduced, Hi-Flo ES filters can reduce dumpster volume by up to 60%.

DATA NOTES:

Standard Hi-Flo ES includes 0.81" (1" nominal) header.

Contact factory for lead times.

CFM value is at a velocity of 500 feet per minute. Filter may be operated at velocities of 350 fpm to 600 fpm without contacting factory.

To establish a schedule for filter change, record initial pressure drop when installed, order filters for change when pressure drop has doubled and service the unit when replacements are available. Maximum recommended pressure drop for this product is 1.5 inches w.g.

The Hi-Flo ES is classified by Underwriters Laboratories as UL Class 900. Maximum operating temperature is 158° F (70° C). Performance tolerances conform to *Section 7.4 of AHRI Standard 850-78*.

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camfil ISO 9001:2008 Certified Quality System

ENERGY COST INDEX

U.S. GREEN BUILDING COUNCIL MEMBER

Camfil USA | 1 North Corporate Drive, Riverdale, NJ 07457 | Tel: (973) 616-7300

Camfil Hi-Flo® ES Model Selection

HFES	/	MV14	/	24	/	24	/	22	/	8	/	
↑		↑		↑		↑		↑		↑		↑
HFES = Hi-Flo ES		Efficiency MV15 = MERV 15 MV14 = MERV 14 MV13 = MERV 13 MV11 = MERV 11		Height (nominal)		Width (nominal)		Depth (nominal)		Number Of Pockets		Options Normally Blank Consult factory for additional options.

Efficiency	Part Number	Model	Nominal size H X W	Pocket Depth	Actual dimensions (inches) H X W X D	Airflow capacity (cfm)	Initial resistance (inches, w.g.)	Media area (sq. ft.)
MERV 15 15-A	405620A15	HFESMV15/24/24/15/10	24 X 24	15	23.31 x 23.31 x 15	2000	0.81"	48.72
	405620A22	HFESMV15/24/24/22/10	24 X 24	22	23.31 x 23.31 x 22	2000	0.62"	71.45
	405620A30	HFESMV15/24/24/30/10	24 X 24	30	23.31 x 23.31 x 30	2000	0.53"	97.03
	405620B15	HFESMV15/24/20/15/8	24 X 20	15	23.31 x 19.31 x 15	1600	0.81"	38.97
	405620B22	HFESMV15/24/20/22/8	24 X 20	22	23.31 x 19.31 x 22	1600	0.62"	57.16
	405620B30	HFESMV15/24/20/30/8	24 X 20	30	23.31 x 19.31 x 30	1600	0.53"	77.62
	405620C15	HFESMV15/24/12/15/5	24 X 12	15	23.31 x 11.31 x 15	1000	0.81"	24.36
	405620C22	HFESMV15/24/12/22/5	24 X 12	22	23.31 x 11.31 x 22	1000	0.62"	35.73
	405620C30	HFESMV15/24/12/30/5	24 X 12	30	23.31 x 11.31 x 30	1000	0.53"	48.52
	405620D15	HFESMV15/20/20/15/8	20 X 20	15	19.31 x 19.31 x 15	1320	0.81"	32.15
	405620D22	HFESMV15/20/20/22/8	20 X 20	22	19.31 x 19.31 x 22	1320	0.62"	47.16
	405620D30	HFESMV15/20/20/30/8	20 X 20	30	19.31 x 19.31 x 30	1320	0.53"	63.98
	405620E15	HFESMV15/20/24/15/10	20 X 24	15	19.31 x 23.31 x 15	1600	0.81"	38.97
	405620E22	HFESMV15/20/24/22/10	20 X 24	22	19.31 x 23.31 x 22	1600	0.62"	57.16
	405620E30	HFESMV15/20/24/30/10	20 X 24	30	19.31 x 23.31 x 30	1600	0.53"	77.62
	405620F15	HFESMV15/12/24/15/10	12 X 24	15	11.31 x 23.31 x 15	1000	0.81"	24.36
	405620F22	HFESMV15/12/24/22/10	12 X 24	22	11.31 x 23.31 x 22	1000	0.62"	35.73
	405620F30	HFESMV15/12/24/30/10	12 X 24	30	11.31 x 23.31 x 30	1000	0.53"	48.52
MERV 14 14A	405619A12	HFESMV14/24/24/12/10	24 x 24	12	23.31 x 23.31 x 12	2000	0.67"	39.21
	405619A15	HFESMV14/24/24/15/10	24 X 24	15	23.31 x 23.31 x 15	2000	0.55"	48.72
	405619A22	HFESMV14/24/24/22/10	24 X 24	22	23.31 x 23.31 x 22	2000	0.45"	71.45
	405619A30	HFESMV14/24/24/30/10	24 X 24	30	23.31 x 23.31 x 30	2000	0.41"	97.03
	405619B12	HFESMV14/24/20/12/8	24 X 20	12	23.31 x 19.31 x 12	1600	0.67"	31.37
	405619B15	HFESMV14/24/20/15/8	24 X 20	15	23.31 x 19.31 x 15	1600	0.55"	39.2125
	405619B22	HFESMV14/24/20/22/8	24 X 20	22	23.31 x 19.31 x 22	1600	0.45"	57.16
	405619B30	HFESMV14/24/20/30/8	24 X 20	30	23.31 x 19.31 x 30	1600	0.41"	77.62
	405619C12	HFESMV14/24/12/12/5	24 X 12	12	23.31 x 11.31 x 12	1000	0.67"	19.61
	405619C15	HFESMV14/24/12/15/5	24 X 12	15	23.31 x 11.31 x 15	1000	0.55"	24.5125
	405619C22	HFESMV14/24/12/22/5	24 X 12	22	23.31 x 11.31 x 22	1000	0.45"	35.73
	405619C30	HFESMV14/24/12/30/5	24 X 12	30	23.31 x 11.31 x 30	1000	0.41"	48.52
	405619D12	HFESMV14/20/20/12/8	20 X 20	12	19.31 x 19.31 x 12	1320	0.67"	25.81
	405619D15	HFESMV14/20/20/15/8	20 X 20	15	19.31 x 19.31 x 15	1320	0.55"	32.2625
	405619D22	HFESMV14/20/20/22/8	20 X 20	22	19.31 x 19.31 x 22	1320	0.45"	47.16
	405619D30	HFESMV14/20/20/30/8	20 X 20	30	19.31 x 19.31 x 30	1320	0.41"	63.98
	405619E12	HFESMV14/20/24/12/10	20 X 24	12	19.31 x 23.31 x 15	1600	0.67"	31.37
	405619E15	HFESMV14/20/24/15/10	20 X 24	15	19.31 x 23.31 x 22	1600	0.55"	39.2125
	405619E22	HFESMV14/20/24/22/10	20 X 24	22	19.31 x 23.31 x 30	1600	0.45"	57.16
	405619E30	HFESMV14/20/24/30/10	20 X 24	30	19.31 x 23.31 x 30	1600	0.41"	77.62
	405619F12	HFESMV14/12/24/12/10	12 X 24	12	11.31 x 23.31 x 22	1000	0.67"	19.61
	405619F15	HFESMV14/12/24/15/10	12 X 24	15	11.31 x 23.31 x 30	1000	0.55"	24.5125
	405619F22	HFESMV14/12/24/22/10	12 X 24	22	11.31 x 23.31 x 22	1000	0.45"	35.73
	405619F30	HFESMV14/12/24/30/10	12 X 24	30	11.31 x 23.31 x 30	1000	0.41"	48.52

See data notes on last page.

PERFORMANCE DATA (continued)

Efficiency	Part Number	Model	Nominal size H X W	Pocket Depth	Actual dimensions (inches) H X W X D	Airflow capacity (cfm)	Initial resistance (inches, w.g.)	Media area (sq. ft.)
MERV 13 13-A	405618A12	HFESMV13/24/24/12/10	24 x 24	12	23.31 x 23.31 x 12	2000	0.52"	39.21
	405618A15	HFESMV13/24/24/15/10	24 X 24	15	23.31 x 23.31 x 15	2000	0.48"	48.72
	405618A22	HFESMV13/24/24/22/10	24 X 24	22	23.31 x 23.31 x 22	2000	0.38"	71.45
	405618A30	HFESMV13/24/24/30/10	24 X 24	30	23.31 x 23.31 x 30	2000	0.34"	97.03
	405618B12	HFESMV13/24/20/12/8	24 X 20	12	23.31 x 19.31 x 12	1600	0.52"	31.37
	405618B15	HFESMV13/24/20/15/8	24 X 20	15	23.31 x 19.31 x 15	1600	0.48"	39.2125
	405618B22	HFESMV13/24/20/22/8	24 X 20	22	23.31 x 19.31 x 22	1600	0.38"	57.16
	405618B30	HFESMV13/24/20/30/8	24 X 20	30	23.31 x 19.31 x 30	1600	0.34"	77.62
	405618C12	HFESMV13/24/12/12/5	24 X 12	12	23.31 x 11.31 x 12	1000	0.52"	19.61
	405618C15	HFESMV13/24/12/15/5	24 X 12	15	23.31 x 11.31 x 15	1000	0.48"	24.5125
	405618C22	HFESMV13/24/12/22/5	24 X 12	22	23.31 x 11.31 x 22	1000	0.38"	35.73
	405618C30	HFESMV13/24/12/30/5	24 X 12	30	23.31 x 11.31 x 30	1000	0.34"	48.52
	405618D12	HFESMV13/20/20/12/8	20 X 20	12	19.31 x 19.31 x 12	1320	0.52"	25.81
	405618D15	HFESMV13/20/20/15/8	20 X 20	15	19.31 x 19.31 x 15	1320	0.48"	32.2625
	405618D22	HFESMV13/20/20/22/8	20 X 20	22	19.31 x 19.31 x 22	1320	0.38"	47.16
	405618D30	HFESMV13/20/20/30/8	20 X 20	30	19.31 x 19.31 x 30	1320	0.34"	63.98
	405618E12	HFESMV13/20/24/12/10	20 X 24	12	19.31 x 23.31 x 15	1600	0.52"	31.37
	405618E15	HFESMV13/20/24/15/10	20 X 24	15	19.31 x 23.31 x 22	1600	0.48"	39.2125
	405618E22	HFESMV13/20/24/22/10	20 X 24	22	19.31 x 23.31 x 30	1600	0.38"	57.16
	405618E30	HFESMV13/20/24/30/10	20 X 24	30	19.31 x 23.31 x 30	1600	0.34"	77.62
	405618F12	HFESMV13/12/24/12/10	12 X 24	12	11.31 x 23.31 x 22	1000	0.52"	19.61
	405618F15	HFESMV13/12/24/15/10	12 X 24	15	11.31 x 23.31 x 30	1000	0.48"	24.5125
	405618F22	HFESMV13/12/24/22/10	12 X 24	22	11.31 x 23.31 x 22	1000	0.38"	35.73
	405618F30	HFESMV13/12/24/30/10	12 X 24	30	11.31 x 23.31 x 30	1000	0.34"	48.52
MERV 11 11-A	405617A12	HFESMV11/24/24/12/10	24 x 24	12	23.31 x 23.31 x 12	2000	0.28"	39.21
	405617A15	HFESMV11/24/24/15/10	24 X 24	15	23.31 x 23.31 x 15	2000	0.26"	48.72
	405617A22	HFESMV11/24/24/22/10	24 X 24	22	23.31 x 23.31 x 22	2000	0.24"	71.45
	405617A30	HFESMV11/24/24/30/10	24 X 24	30	23.31 x 23.31 x 30	2000	0.23"	97.03
	405617B12	HFESMV11/24/20/12/8	24 X 20	12	23.31 x 19.31 x 12	1600	0.28"	31.37
	405617B15	HFESMV11/24/20/15/8	24 X 20	15	23.31 x 19.31 x 15	1600	0.26"	39.2125
	405617B22	HFESMV11/24/20/22/8	24 X 20	22	23.31 x 19.31 x 22	1600	0.24"	57.16
	405617B30	HFESMV11/24/20/30/8	24 X 20	30	23.31 x 19.31 x 30	1600	0.23"	77.62
	405617C12	HFESMV11/24/12/12/5	24 X 12	12	23.31 x 11.31 x 12	1000	0.28"	19.61
	405617C15	HFESMV11/24/12/15/5	24 X 12	15	23.31 x 11.31 x 15	1000	0.26"	24.5125
	405617C22	HFESMV11/24/12/22/5	24 X 12	22	23.31 x 11.31 x 22	1000	0.24"	35.73
	405617C30	HFESMV11/24/12/30/5	24 X 12	30	23.31 x 11.31 x 30	1000	0.23"	48.52
	405617D12	HFESMV11/20/20/12/8	20 X 20	12	19.31 x 19.31 x 12	1320	0.28"	25.81
	405617D15	HFESMV11/20/20/15/8	20 X 20	15	19.31 x 19.31 x 15	1320	0.26"	32.2625
	405617D22	HFESMV11/20/20/22/8	20 X 20	22	19.31 x 19.31 x 22	1320	0.24"	47.16
	405617D30	HFESMV11/20/20/30/8	20 X 20	30	19.31 x 19.31 x 30	1320	0.23"	63.98
	405617E12	HFESMV11/20/24/12/10	20 X 24	12	19.31 x 23.31 x 15	1600	0.28"	31.37
	405617E15	HFESMV11/20/24/15/10	20 X 24	15	19.31 x 23.31 x 22	1600	0.26"	39.2125
	405617E22	HFESMV11/20/24/22/10	20 X 24	22	19.31 x 23.31 x 30	1600	0.24"	57.16
	405617E30	HFESMV11/20/24/30/10	20 X 24	30	19.31 x 23.31 x 30	1600	0.23"	77.62
	405617F12	HFESMV11/12/24/12/10	12 X 24	12	11.31 x 23.31 x 22	1000	0.28"	19.61
	405617F15	HFESMV11/12/24/15/10	12 X 24	15	11.31 x 23.31 x 30	1000	0.26"	24.5125
	405617F22	HFESMV11/12/24/22/10	12 X 24	22	11.31 x 23.31 x 22	1000	0.24"	35.73
	405617F30	HFESMV11/12/24/30/10	12 X 24	30	11.31 x 23.31 x 30	1000	0.23"	48.52

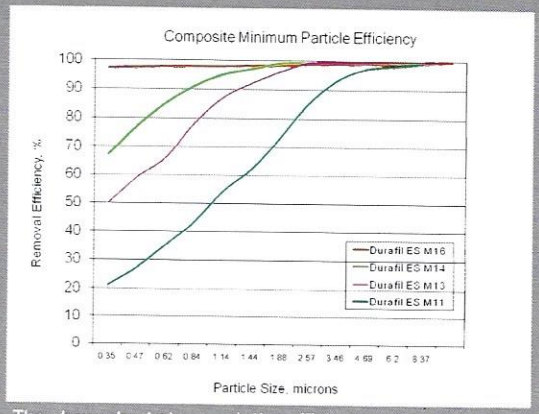
See data notes on last page.

camfil

Durafil® ES²

High Efficiency, High Capacity, Energy Saving, Mini-Pleated V-Bank Air Filter

Saves 30-35% more energy than any other V-style filter



The above chart shows relative efficiency values at various particle sizes when tested in accordance with ASHRAE Standard 52.2-2007. When tested in accordance with Appendix J of that Standard the Durafil ES maintains these efficiency values throughout the life of the filter.

The Camfil Durafil ES² (Energy Saver) provides high efficiency ASHRAE air filter performance in a compact energy efficient design. The Durafil ES² includes:

- A computer-optimized pleat-to-height ratio creating a radial air exiting and air entering design that offers a 60% larger outlet and 30% larger inlet than any other V-bank style air filter resulting in lower pressure drop and significant energy savings.
- The highest volume of microfine fiber filter media area available for higher dust holding capacity, longer life and lower average pressure drop over the life of the filter to save energy.
- A special grade of energy saving media with engineered characteristics to reduce pressure drop.
- Is available in four standard efficiencies — MERV 11, MERV 13, MERV 14 and MERV 16 per ASHRAE Standard 52.2. The Durafil ES² has a MERV-A value of 11, 13, 14 and 16 respectively when tested using the conditioning step as specified in Appendix J of the same Standard.
- Includes an integral prefilter spacer section designed to minimize filtration system static pressure when a prefilter is positioned on the face of the Durafil ES². The lives of the final filter and prefilter are extended and pressure drop is minimized to save energy.
- Includes media separators creating uniform airflow throughout the media pack.
- Incorporates a unique sealant channel ensuring media pack-to-frame bonding to prevent air bypass.
- Includes an impact-resistant plastic enclosing frame with plastic media pack supports ensuring a rigid and durable filter. Bridging supports on each 'V' provide handles for installation and transport and maintains the rigidity of the filter. The frame also has built-in spring fastener locations, built in prefilter fastener insertion locations and recessed front and back inlet and outlet handles for changing and aligning the filter during installation.
- Includes a one-inch nominal size header as an integral component of the frame for added stability and a secure fit into filter holding mechanism or housing. The frame includes a cavity for attaching Camfil's C-81 fasteners for face mounting of 30/30® prefilters.
- Includes a sealing gasket on one vertical header to eliminate air bypass between headers in multiple filter systems.
- Can be installed in systems with airflow capacities to 3,000 cfm. Maximum pressure drop capability is guaranteed to 2.0" w.g. and filter integrity is guaranteed to 10.0" w.g.
- Is the lightest weight V-bank filter available.
- Has an ECI[1] value of five stars.

The Durafil's superior performance characteristics relating to human and environmental health, energy efficiency, materials selection and indoor environmental quality make it the final filter of choice for those facilities pursuing green building status.

Durafil® ES²

High Efficiency, High Capacity, Energy Saving, Mini-Pleated V-Bank Air Filter

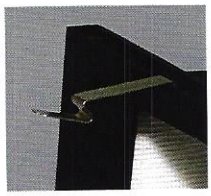
Performance

ASHRAE Efficiency	Part Number	Nominal Depth (inches)	Nominal Size (H x W) (inches)	Actual Depth (inches)	Actual Dimensions H x W (inches)	Initial Resistance (inches w.g.)	Airflow Capacity (cfm)	Media Area (ft²)
MERV 16 [a] MERV 16-A [b]	855080-014	12	24 x 24	11.63	23.38 x 23.38	0.64	2000	200
	855080-030		20 x 24		19.38 x 23.38		1500	160
	855080-021		12 x 24		11.38 x 23.38		1000	100
	855080-188[d]		20 x 20		19.38 x 19.38	0.80	1250	125
MERV 14 [a] MERV 14-A [b]	855080-009	12	24 x 24	11.63	23.38 x 23.38	0.29	2000	200
	855080-006		20 x 24		19.38 x 23.38		1500	160
	855080-003		12 x 24		11.38 x 23.38		1000	100
	855080-065[d]		20 x 20		19.38 x 19.38	0.33	1250	125
MERV 13 [c] MERV 13-A [b]	855080-008	12	24 x 24	11.63	23.38 x 23.38	0.27	2000	200
	855080-005		20 x 24		19.38 x 23.38		1500	160
	855080-002		12 x 24		11.38 x 23.38		1000	100
	855080-066[d]		20 x 20		19.38 x 19.38	0.32	1250	125
MERV 11 MERV 11-A [b]	855080-007	12	24 x 24	11.63	23.38 x 23.38	0.22	2000	200
	855080-004		20 x 24		19.38 x 23.38		1500	160
	855080-001		12 x 24		11.38 x 23.38		1000	100
	855080-063[d]		20 x 20		19.38 x 19.38	0.27	1250	125

DATA NOTES:

[a] May provide additional LEED credits.
[b] Discharged efficiency per appendix J of ASHRAE Standard 52.2.
[c] Minimum efficiency selection for LEED consideration.
[d] 20" by 20" size does not have spacing plenum for prefilter application, contact factory for prefiltration guidance. 20" by 20" size does not have radial configuration.
Airflow may be in either direction.
Schedule air filters for change when initial pressure drop has doubled. Final pressure drop should not exceed 1.50" w.g.
The Durafil ES² is listed UL 900 by Underwriters Labortories.
Maximum continuous operating temperature 175° F. (79° C.), relative humidity 99%.
U.S.Patent No. 6,447,566. Performance tolerance in accordance with ARI Standard 850.
Options: Available with gaskets in any location. Available with dual headers as shown to the right. See Product Sheet 1515B.

ESB Model has dual headers.



C-81 fasteners facilitate prefilter face mounting.

1.0 General

1.1 · Air filters shall be V-Bank mini-pleat fiberglass disposable type with pleat separators, polyurethane pack-to frame sealant, polystyrene enclosing frame and have an ECI value of five stars.

1.2 · Sizes shall be as noted on drawings or other supporting materials.

2.0 Construction

2.1 · Filter media shall be of microfine glass fibers formed into uniform pleats with a spacing of 8.5 pleats per inch and a uniform pleat height of 22mm. Pleats shall be separated at 25mm intervals to ensure pleat separation and uniform airflow through the filter pack.

2.2 · Pleats media packs shall be assembled into a V-bank configuration with sufficient total media area to meet airflow requirements. The filter outlet shall be radial in shape with a maximum of 60% open area to maintain low-pressure drop and uniform airflow (20" by 20" shall be straight V-style design).

2.3 · The media packs shall be bonded to the inside periphery of an ABS enclosing frame with a polyurethane sealant. The enclosing frame shall include top and bottom molded tracks as in integral part of the frame to ensure a proper seal.

2.4 · Media packs shall be recessed at least 1" from the air entering side of the enclosing frame to allow uniform airflow when a prefilter is mounted directly to the enclosing frame.

2.5 · Rigid plastic end caps shall be mechanically fastened to the top and bottom of the media pack enclosing structure to ensure a rigid and durable filter.

2.6 · Carrying handles shall be an integral part of the filter frame and shall bridge from media pack to media pack providing additional filter support and filter rigidity. Handles shall include fastener connection locations for the application of spring mounting fasteners when the filter is applied in reverse flow applications. .

3.0 Performance

3.1 · The filter shall have a Minimum Efficiency Reporting Value of MERV (11, 13, 14, 16) when evaluated under the guidelines of ASHRAE Standard 52.2. It shall also have a MERV-A rating of (11, 13, 14, 16) when evaluated under ASHRAE Standard 52.2, Appendix J.

3.2 · Initial resistance to airflow shall not exceed (0.24, 0.30, 0.32, 0.70) inches w.g. at an airflow of 500 fpm for 24 x 24, 24 x 12 and 24 x 20 sizes. On 20" by 20" respective pressure drops shall be (0.88, 0.36, 0.35, 0.30) inches w.g. at an airflow of 500 fpm.

3.3 · Filter shall be listed UL 900 by Underwriters Laboratories.

3.4 - The filter shall be capable of withstanding 10.00" w.g. without failure of the media pack.

3.5 · Manufacturer shall provide evidence of facility certification to ISO 9001:2008.

3.6 · Filter shall have a 5-Star rating when evaluated per Energy Cost Index (ECI).

Supporting Data · Provide product test reports for each listed efficiency including all details as prescribed in ASHRAE Standards 52.2 including Appendix J.

Filters shall be Camfil Durafil ES or equal.

(Items in parentheses ( ) require selection.

© Copyright Camfil # 1510 · 0916

Blue Heaven Technologies

2820 S. English Station Road - Louisville, KY 40299

Tel: (502) 357-0132 Fax (502) 267-8379

Date: 4-Nov-14 TEST NO. 14-1860ARev1

ASHRAE Standard 52.2-2012
APPENDIX J TEST REPORT
Efficiency with KCL Conditioning

Filter Description

Manufacturer	Camfil USA, Inc.
Filter Model	DU4V-ES-2424-MV13
Part Number	85508008
Generic Filter Type	Durafil ES - V Bed Air Filter
Nominal Dimensions (H x W x D)	24" x 24" x 12"
Pocket / Pleat Quantity	4 V
Media Type	Wet Laid Glass
Est. Gross Media Area	200 Ft²
Adhesive Type	N A

Test Conditions

Conditioning Aerosol Type	KCL		
Loading Dust Type	N A	Test Air Temp (degrees F.)	71
Barometric Pressure (In. Hg.)	29.71	Relative Humidity (%)	40

Test Results

Airflow Rate (CFM)	**1968**
Nominal Face Velocity (fpm)	**492**
Initial Resistance (in WG)	**0.24**
E1-A (%) Min. Composite Efficiency after KCL Conditioning 0.30 - 1.0 um	**69**
E2-A (%) Min. Composite Efficiency after KCL Conditioning 1.0 - 3.0 um	**92**
E3-A (%) Min. Composite Efficiency after KCL Conditioning 3.0 - 10.0 um	**99**
***Minimum Efficiency Reporting Value after KCL Conditioning (MERV-A)**	**MERV 13-A @ 1968 CFM**

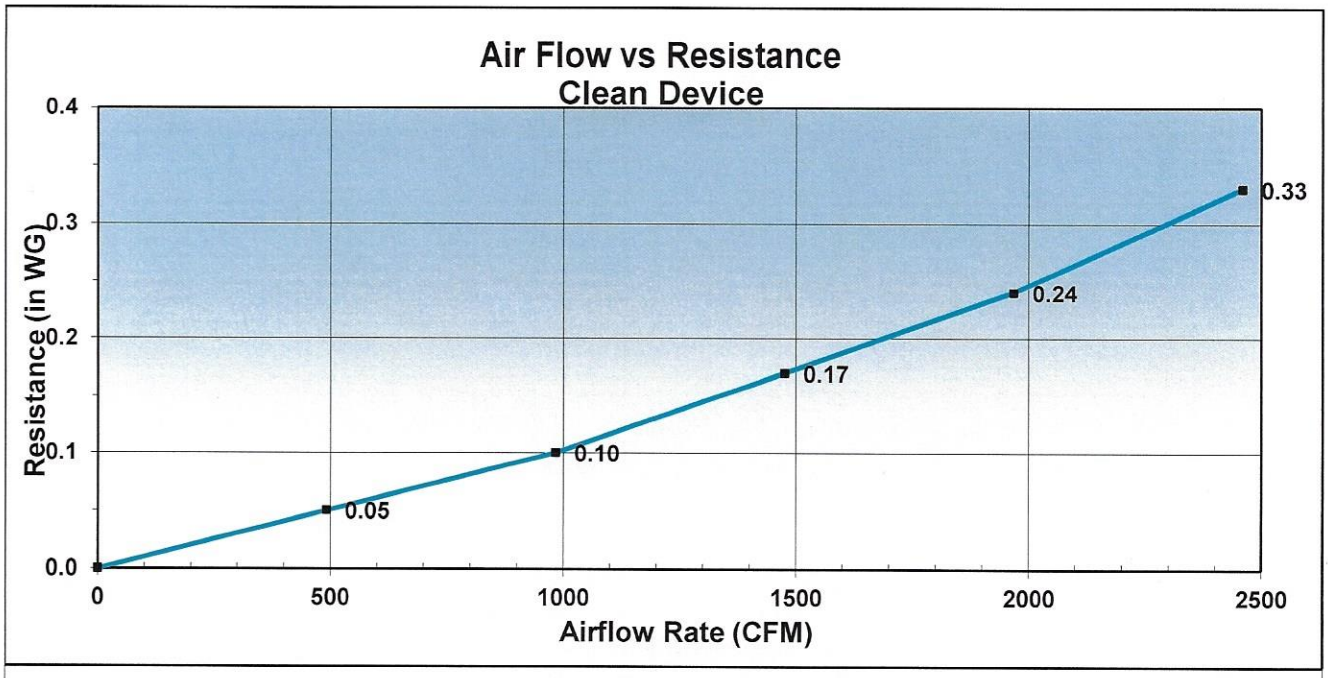
(* assumes no efficiency reduction during subsequent ASHRAE dust loading)

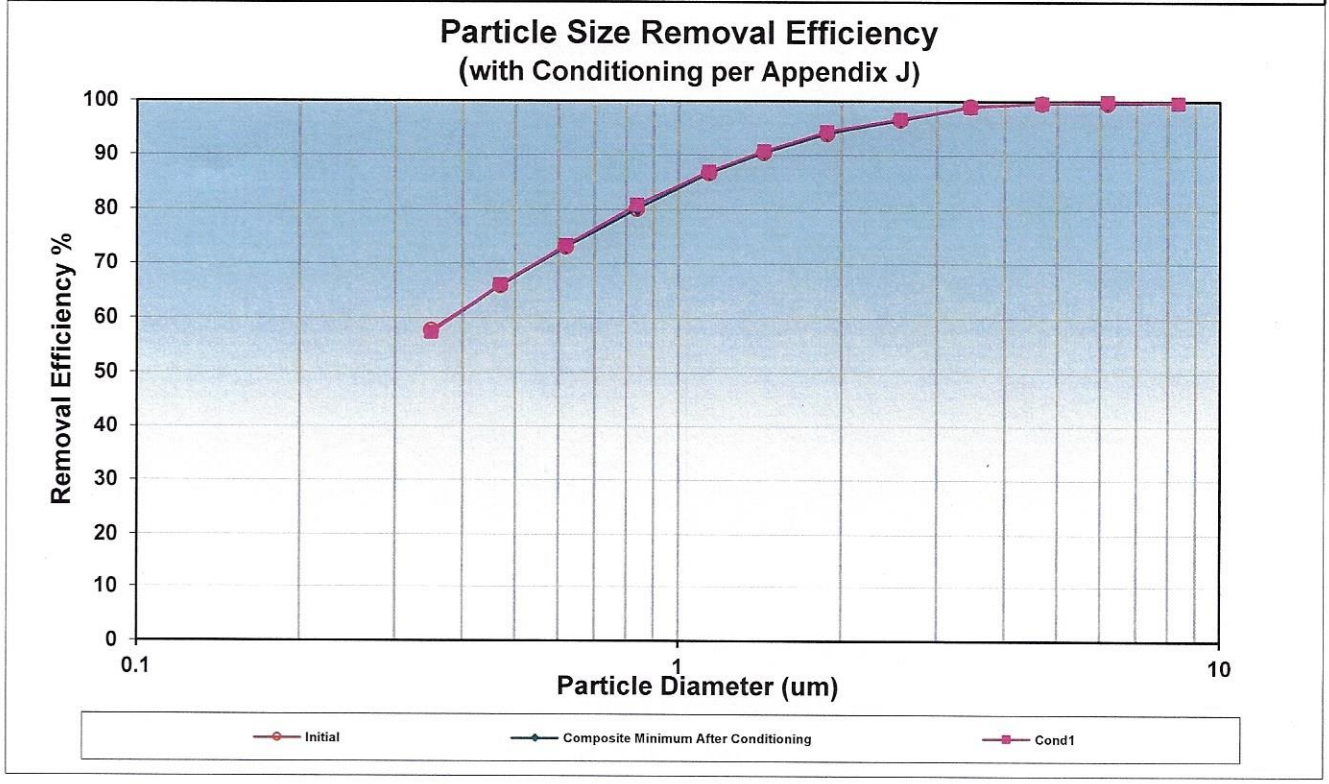
Comments Tested For: Camfil USA, Inc.

Test Performed by: RAS Approved By: Test Completed: 4-Nov-14

Blue Heaven Technologies
2820 S. English Station Rd. Louisville, KY
502 357 0132

Test No. 14-1860ARev1
Date: 04-Nov-14





Blue Heaven Technologies
2820 S. English Station Rd. Louisville, KY
502 357 0132

Data - Initial Resistance

Airflow (CFM)	Resistance (in WG)
0	0.00
492	0.05
984	0.10
1476	0.17
1968	0.24
2460	0.33

Data - Particle Removal Efficiency per Appendix J

Particle Size Range (um)	Geometric Mean Diam (um)	Particle Removal Efficiency (%)		
0.30 - 0.40	0.35	57.3	57.6	57.3
0.40 - 0.55	0.47	65.8	65.8	66.0
0.55 - 0.70	0.62	73.0	73.0	73.3
0.70 - 1.00	0.84	80.0	80.0	80.7
1.00 - 1.30	1.14	86.6	86.6	86.9
1.30 - 1.60	1.44	90.3	90.3	90.7
1.60 - 2.20	1.88	93.9	93.9	94.3
2.20 - 3.00	2.57	96.5	96.5	96.7
3.00 - 4.00	3.46	98.9	99.0	98.9
4.00 - 5.50	4.69	99.6	99.6	99.6
5.50 - 7.00	6.20	99.6	99.6	99.9
7.00 - 10.00	8.37	99.7	100.1	99.7
		Composite Min Eff (CME)	Initial	Conditioning Increment 1

Conditioning Procedure Details

(included for reference as required by Appendix J11.1)

	Initial	Cond 1					Post	Req'd by App J
Background concentration (particles cm-3) ----->	16.59						11.22	low
Correlation ratio small to large particles ----->	33,397						30,259	> 20,000
Avg conditioning aerosol concentration (particles cm-3) ----->		119,052						< 1,000,000
Incremental conditioning duration (minutes) ----->		484.0						record
Cumulative conditioning duration (minutes) ----->		484.0						
Incremental CT (particle cm-3 min) ----->		64,288,032						> 6.4 x 10E7
Cumulative conditioning CT (particles cm-3 min) ----->		64,288,032						< 1.2 x 10E9
Dust Holding Capacity DHC-A (grams) ----->		NA						record

RTI INTERNATIONAL
3040 Cornwallis Rd. PO Box 12194 Research Triangle Park, NC 27709
Telephone 919 541 6941 Fax 919 541 6936 www.rti.org

ASHRAE 52.2-2012 WITH APPENDIX J CONDITIONING TEST REPORT

Manufacturer: Camfil
Product Name: Durafil ES2 MV14

RTI Report No. BY07011602

Test Laboratory:
RTI
3040 Cornwallis Road
Research Triangle Park, NC 27709
919-541-6941
mko@rti.org

ASHRAE Std. 52.2-2012 with Appendix J Conditioning Report Summary

This report applies to the tested device only.

Laboratory Data

RTI Report No.	BY07011602		Date	1-Jul-16	
Test Laboratory	Research Triangle Institute				
Operator	Pope		Supervisor	Owen/Hanley	
Particle Counter(s):	Brand	Climet	Model	500 Spectrometer	

Device Manufacturer's Data

Manufacturer	Camfil			
Product Name	Durafil ES2 MV14			
Product Model				
Test requested by	Camfil			
Sample obtained from	Camfil			
Catalog rating:	Airflow rate	NA	Initial dP (in. wg)	NA
Specified test conditions:	Airflow (cfm)	1968	Final dP (in. wg)	1.50
	Face Velocity (fpm)	492		

Device Description

Nominal Dimensions (in.):	24 x 24 x 12	(height x width x depth)	
Generic name	V-cell	Media color	white
Amount and type of adhesive	NA		
Other attributes	4 Vs		

Test Conditions

Airflow (cfm)	1968	Temperature (F)	77	RH (%)	45
Face Velocity (fpm)	492	Final Pressure Drop (in. wg)		1.50	
Test aerosol type:	KCl				
Remarks	This filter was conditioned with high concentration, sub 0.1 µm KCl aerosol per Appendix J.				

Resistance Test Results

Initial resistance (in. wg)	0.34	Final resistance (in. wg)	1.50

Minimum Efficiency Reporting Data

Composite average efficiencies	E1-A	76-A	E2-A	96-A	E3-A	99-A
Minimum efficiency reporting value (MERV-A):		14-A	@	1968 cfm		
Air cleaner average Arrestance (A_{AVG}-A):		100%				
Dust Holding Capacity (DHC-A):		390.9 g				

TABULATED DATA SUMMARY
Report No. BY07011602
Research Triangle Institute

Summary of Test Conditions:

Product Manufacturer	Camfil
Product Name	Durafil ES2 MV14
Nominal Dimensions (in.)	24 x 24 x 12
Airflow (cfm)	1968
Final Resistance (in. H2O)	1.50

		Efficiency (%) per Indicated Size Range											
OPC Channel Number		1	2	3	4	5	6	7	8	9	10	11	12
Min. Diam. (µm)		0.3	0.4	0.55	0.7	1	1.3	1.6	2.2	3	4	5.5	7
Max. Diam. (µm)		0.4	0.55	0.7	1	1.3	1.6	2.2	3	4	5.5	7	10
Geo. Mean Diam (µm)		0.35	0.47	0.62	0.84	1.14	1.44	1.88	2.57	3.46	4.69	6.20	8.37
	Run No.												
Initial efficiency	BY07061601	65	73	80	87	93	96	97	98	99	99	100	99
after first cond.	BY07071601	66	74	81	88	93	96	97	98	99	99	99	99
after 25% dust load	BY07071602	93	96	97	98	98	98	99	99	99	100	100	100
after 50% dust load	BY07081601	96	96	97	98	98	98	99	99	99	100	100	100
after 75% dust load	BY07081602	96	97	97	97	98	98	98	99	99	100	100	100
after final dust load	BY07081603	96	96	97	97	98	98	99	100	99	100	101	101
Minimum Composite Efficiency		65	73	80	87	93	96	97	98	99	99	99	99

E1-A = 76 -A
E2-A = 96 -A
E3-A = 99 -A

MERV-A = 14 -A

Resistance to Airflow:

Airflow (%)	Airflow (m3/s)	Airflow (cfm)	Air Velocity (fpm)	Air Velocity (m/s)	Resistance (in. H2O)	Resistance (Pa)
50	0.464	984	246	1.250	0.13	33
75	0.697	1476	369	1.875	0.22	56
100	0.929	1968	492	2.499	0.34	85
125	1.161	2460	615	3.124	0.47	117

Conditioning Steps

	1st
Concentration #/cc*	3.5E+05
Time (min)	376
Exposure #/cc-min*	1.3E+08

* Approximate

Resistance to Airflow with Loading at 1968 cfm

	Resistance (in. H2O)	Resistance (Pa)	Dust Fed (g)
Initial efficiency	0.34	85	
after first cond.	0.34	85	
after 25% dust load	0.63	157	179.4
after 50% dust load	0.92	229	80.1
after 75% dust load	1.21	301	67.8
after final dust load	1.50	373	64.0
Dust Holding Capacity (DHC-A)			390.9 g
Arrestance (A_{AVG}-A)			100%

TABULATED DATA SUMMARY
Report No. BY07011602
Research Triangle Institute

Summary of Test Conditions:

Product Manufacturer	Camfil
Product Name	Durafil ES2 MV14
Nominal Dimensions (in.)	24 x 24 x 12
Airflow (cfm)	1968

Release Rate (%) per Indicated Size Range

OPC Channel Number	1	2	3	4	5	6	7	8	9	10	11	12
Min. Diam. (µm)	0.3	0.4	0.55	0.7	1	1.3	1.6	2.2	3	4	5.5	7
Max. Diam. (µm)	0.4	0.55	0.7	1	1.3	1.6	2.2	3	4	5.5	7	10
Geo. Mean Diam (µm)	0.35	0.47	0.62	0.84	1.14	1.44	1.88	2.57	3.46	4.69	6.20	8.37
after first dust load	0	0	0	0	0	0	0	0	0	0	0	0
after second dust load	0	0	0	0	0	0	0	0	0	1	1	2
after third dust load	0	0	0	0	0	0	0	0	0	1	1	2
after fourth dust load	0	0	0	0	0	0	0	0	0	1	1	2

INVITATION FOR BID (IFB) #041-21
HVAC FILTERS and BELTS
DETAILED BID SPECIFICATIONS

SECTION FIVE
HVAC BELTS

1) All belts must be manufactured in the USA.

2) All orders must be packaged and labeled per facility and per equipment number (no bulk packaging).

3) All orders are to be set up for auto ship per Douglas Counties Maintenance Schedule.

4) No minimum order requirements.

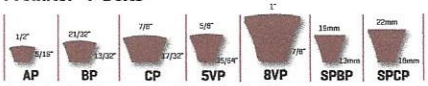
5) Belt list is provided, attached and incorporated herein as Exhibit A, and includes estimated annual quantities.

BELT I.D. CHART

For comprehensive product information see Gates Industrial Power Transmission Systems Catalog No. 13993

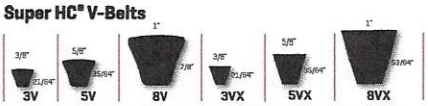
Single V-Belts

Predator® V-Belts



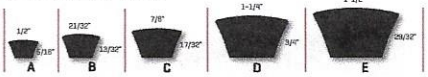
Aramid tensile cords provide superior service on high impact, shock-loaded drives.

Super HC® V-Belts



Combine cross section designation with Outside Circumference (O.C.) to the nearest whole number, plus a zero to determine Belt Part Number. Example: 5/8" top width 5VX belt with 80" O.C. equals 5VX800 V-Belt. X designates molded notch construction.

Hi-Power® II V-Belts



Combine cross section plus Inside Circumference (I.C.) to determine Belt Part Number. To calculate I.C., subtract the following values from the O.C.:

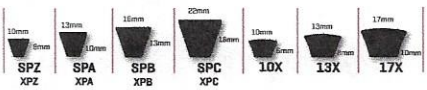
Cross Section	A	B	C	D	E
Subtract from O.C.	2"	3" (Above 210", 1.0")	4" (Above 210", 2.0")	5" (Above 210", 3.0")	7" (Above 210", 4.0")

Dubl-V belts are available in AA, BB, CC and DD cross sections.

Tri-Power® V-Belts

Tri-Power construction is identified by its distinctive molded notch configuration.

Metric Power™ V-Belts



Molded notch construction is available in lengths under 3,000mm only.

Multi-Speed Belts

First two digits indicate top width in sixteenths of an inch. Next two digits designate sheave angle, in degrees, that the belt is designed to fit. Last three or four digits indicate pitch length to the nearest tenth of an inch.

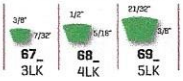
Example: 2326V310 Belt

23	26	V	310
Top Width in 16ths of an inch: 23/16" = 1-7/16"	Sheave Angle in Degrees (26)	Multi-Speed	Pitch Circumference to the nearest 10th inch: 31.0"

Truflex® & PoweRated® Light Duty V-Belts

Truflex®
Part numbers are derived from industry standard number. First digit in Gates number corresponds to first two digits of industry standard number. Belt length is to the nearest tenth of an inch.
Example: 2450 = 4L450 Belt has 1/2" top width and 45.0" O.C.

PoweRated®
First two digits indicate belt top width.
67 = 3/8", **68** = 1/2", **69** = 21/32"
Last two digits indicate length in inches.
Example: 6823 Belt has a 1/2" top width and a 23" O.C.



PowerBand® Joined Belts

Made by joining two or more single V-Belts with a permanent, high strength tie-band. PowerBand belts prevent turn-over or jumping off the sheave problems associated with heavily shock loaded drives using individual belts. PowerBand belts use the same system of size and length designation as individual belts.

Predator® PowerBand® Belts

Available in CP, 3VP, 5VP and 8VP sections.

Super HC® & Super HC® Molded Notch PowerBand® Belts

Available in 3V, 3VX, 5V, 5VX and 8V sections.

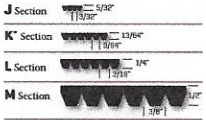
Hi-Power® II PowerBand® Belts

Available in A, B, C and D sections.

Micro-V® Belts

Identified by a three-part system consisting of:
[1] A standard length designation
[2] Cross Section
[3] Number of ribs

Example: 780J12 Belt
[1] An effective length of 78"
[2] J cross section
[3] 12 ribs wide

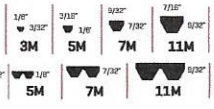


*Automotive Product

Polyflex® and Polyflex® JB® Belts

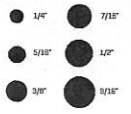
Identified by a three-part system consisting of:
[1] Number of Strands
[2] Top width of belt in mm
[3] Length in mm

Example: 3/7M850 Belt:
[1] 3 strands
[2] 7M profile
[3] An effective length of 850mm



Round Belts

Round Endless



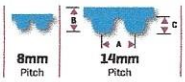
Power Round™ Heavy-Duty Construction

Synchronous Belts

All synchronous belts are identified in a similar manner, in either English or metric units. Belts are identified by:
1. Pitch: Distance in inches or millimeters between two adjacent tooth centers as measured on the belt pitch line.
2. Pitch Length: Total length (circumference) in inches or millimeters as measured along the pitch line. It is equal to the pitch multiplied by the number of teeth in the belt.
3. Width: Always shown as the last part of the size designation. Denotes width in inches or millimeters.

Poly Chain® GT® Carbon™ Belts

Examples: 8MGT-640-12 Belt 8MX-22S-12 GT2 Sprocket



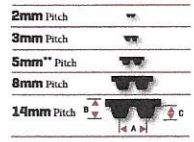
	Standard Widths (mm)	A	B	C
8mm	12, 21	8mm .315 in.	5.9mm .233 in.	3.4mm .135 in.
	36, 62			
14mm	20, 37	14mm .551 in.	10.2mm .401 in.	6mm .236 in.
	68, 90, 125			

Note: Gates recommends that Poly Chain® GT® Carbon™ belts run only in Poly Chain® GT®2 sprockets when used for new applications. Gates recommends that Poly Chain® GT® Carbon™ belts be used for replacement belts in Poly Chain® GT®2 sprockets.

PowerGrip® GT®2 Belts

Examples: 640-8MGT-20 Belt | 5MR-300-09 Belt
P22-8MGT-20 1108 Sprocket | P18-5MGT-15 Sprocket

	Standard Widths (mm)	A	B	C
2mm	4, 6, 9	2mm .0787 in.	1.52mm .060 in.	.76mm .030 in.
3mm	6, 9, 15	3mm .1181 in.	2.41mm .095 in.	1.14mm .045 in.
5mm	9, 15, 25	5mm .197 in.	3.81mm .150 in.	1.93mm .076 in.
8mm	12, 20, 30 50, 85	8mm .315 in.	5.59mm .220 in.	3.28mm .129 in.
14mm	40, 55, 85 115, 170	14mm .551 in.	9.91mm .390 in.	5.84mm .230 in.



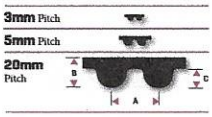
Note: PowerGrip® GT®2 belts must be used with PowerGrip® GT®2 sprockets for new designs.
Note: 8mm and 14mm pitch PowerGrip® GT®2 belts can be used to replace non-Gates curvilinear belts in most instances. Reference gates.com/letterchange for specific interchange information.

**5mm pitch also available in Poly Chain® construction. See Catalog No. 13993.

PowerGrip® HTD® Belts

Examples: 350-5M-15 Belt P28-5M-15AL Sprocket

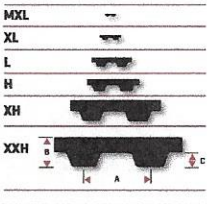
	Standard Widths (in.)	A	B	C
3mm	6, 9, 15	3mm .1181 in.	2.41mm .095 in.	1.22mm .048 in.
5mm	9, 15, 25	5mm .197 in.	3.81mm .150 in.	2.08mm .082 in.
20mm	115, 170 230, 290 340	20mm .787 in.	13.2mm .520 in.	8.4mm .330 in.



PowerGrip® Timing Belts

Examples: 210L100 Belt 7L18L100 Pulley

	Standard Widths (in.)	A	B	C
MXL	1/8, 3/16 1/4	2.03mm .080 in.	1.14mm .045 in.	0.51mm .020 in.
XL	1/4, 3/8	5.08mm .200 in.	2.29mm .090 in.	1.27mm .050 in.
L	1/2, 3/4 1	9.53mm .375 in.	3.60mm .140 in.	1.91mm .075 in.
H	3/4, 1 1-1/2, 2, 3	12.70mm .500 in.	4.10mm .160 in.	2.29mm .090 in.
XH	2, 3, 4	22.23mm .875 in.	11.20mm .440 in.	6.35mm .250 in.
XXH	2, 3, 4, 5	31.75mm 1.250 in.	15.70mm .820 in.	9.53mm .375 in.

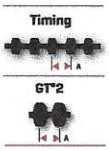


PowerGrip® Twin Power® Belts

Example: TP800H100 Belt

Twin Power® PowerGrip® Timing Belts

	Standard Widths (mm)	A
XL	1/4, 3/8	5.08mm .200 in.
L	1/2, 3/4 1	9.53mm .375 in.
H	3/4, 1 1-1/2, 2, 3	12.70mm .500 in.



Note: Additional sizes available.

Example: TP1200-8MGT-20 Belt (GT2)

Twin Power® PowerGrip® GT®2 Belts

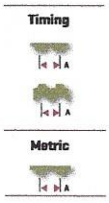
	Standard Widths (mm)	A
3mm	6, 9, 15	3mm .118 in.
5mm	9, 15, 25	5mm .197 in.
8mm	20, 30, 50 85	8mm .315 in.
14mm	40, 55, 85 115, 170	14mm .552 in.

Synchro-Power® Polyurethane Belts

Example: 15-270-8

Synchro-Power® Belts

	A
T2.5	2.5mm .098 in.
T5	5mm .197 in.
T10	10mm .394 in.
AT5	5mm .197 in.
AT10	10mm .394 in.
DT5	5mm .197 in.
DT10	10mm .394 in.



Example: 270H075U

Synchro-Power® Timing Belts

	A
MXL	2.03mm .080 in.
XL	5.08mm .200 in.
L	9.53mm .374 in.
H	12.70mm .500 in.

Long-Length Belting

PowerGrip® and Poly Chain® Belting

Poly Chain GT Carbon (Carbon)	8mm and 14mm
PowerGrip GT2 (Fiberglass)	2mm, 3mm, 5mm and 8mm
PowerGrip HTD (Fiberglass)	3mm, 5mm, 8mm and 14mm
PowerGrip Timing (Fiberglass)	MXL, XL, L and H

PowerGrip GT2 (Steel)	5mm and 8mm
PowerGrip HTD (Steel)	14mm
PowerGrip Timing (Steel)	H

Notes: Minimum order quanity - 50ft. fiberglass construction, 98 ft. steel construction, 50 ft. Poly Chain GT Carbon construction. Refer to Catalog No. 13993 for widths.

Gates Mectrol® Polyurethane Belting

Available in: T5, AT5, T10, AT10, T20, HTD5, HTD8, HTD14, STD5, STD8, XL, L, H, XH
Available with: Nylon tooth and/or back. Polyurethane, Rubber, Foam, PVC, and Special Backings.
Contact Gates Mectrol 800.394.4844

Warning: Do not use Gates belts, pulleys, or sprockets on any aircraft propeller or motor drive systems or in-flight accessory drives. Gates products are not designed or intended for aircraft use.

www.gates.com/pt

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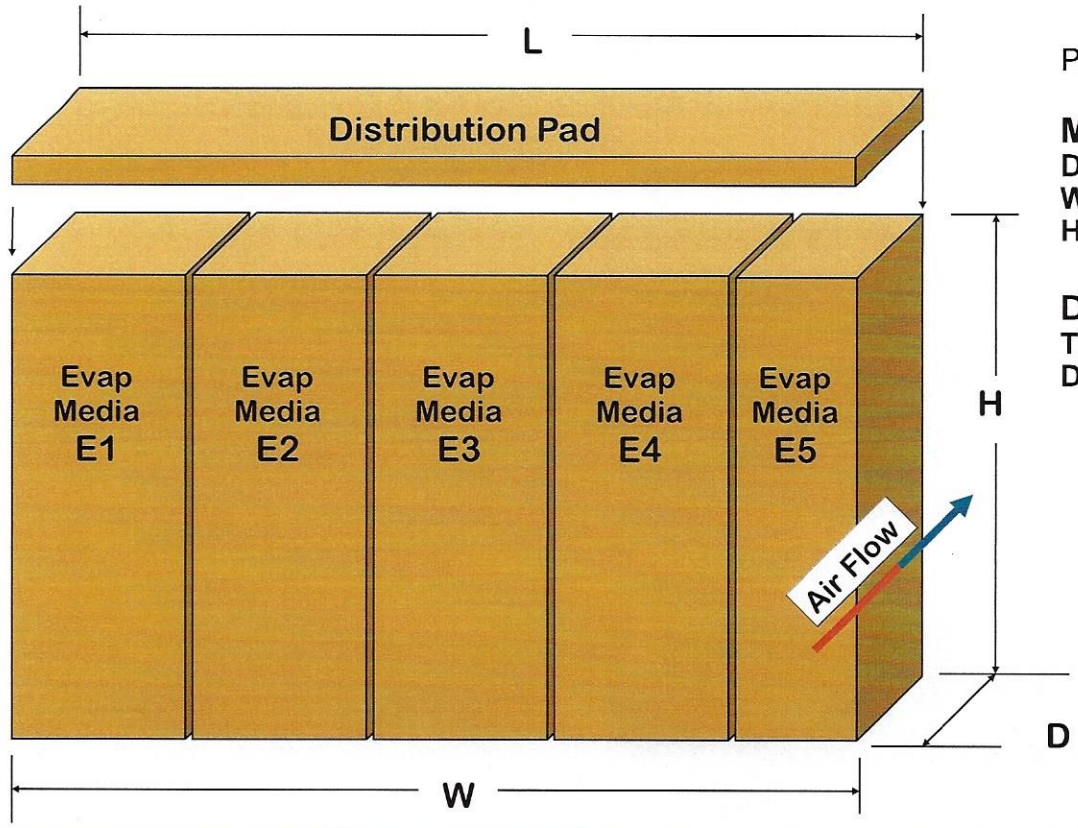
12998-J 5/12

DRIVE SHUTDOWN & THOROUGH INSPECTION

The adjusted belt weights for use with the Gates Sonic Tension Meter are shown in the following table.

Belt Product Family	Belt Cross Section	Belt Type	Adjusted Belt Weight (grams/meter)
Super HC® V-Belts	3VX	Single	53
	5VX	Single	140
	8VX	Single	383
	3V	Single	72
	5V	Single	200
	8V	Single	510
	3VX	PowerBand®	65
	5VX	PowerBand®	157
	3V	PowerBand®	96
	5V	PowerBand®	241
	8V	PowerBand®	579
Super HC® XP™ V-Belts	5VXP	Single	163
	5VXP	PowerBand®	
Predator® V-Belts	5VP	Single	217
	8VP	Single	528
	AP	Single	114
	BP	Single	174
	CP	Single	323
	SPBP	Single	208
	SBCP	Single	375
	3VP	PowerBand®	89
	5VP	PowerBand®	217
	8VP	PowerBand®	528
	BP	PowerBand®	212
	CP	PowerBand®	332
Tri-Power® V-Belts	AX	Single	79
	BX	Single	136
	CX	Single	216
	AX	PowerBand®	100
	BX	PowerBand®	163
	CX	PowerBand®	281
Hi Power® II V-Belts	A	Single	95
	B	Single	168
	C	Single	275
	D	Single	553
	E	Single	965
	A	PowerBand®	151
	B	PowerBand®	200
	C	PowerBand®	342
	D	PowerBand®	663
Hi Power® II Dubl-V Belts	AA	Single	125
	BB	Single	194
	CC	Single	354
	DD	Single	750
Micro-V® Belts	H	Single	5.3
	J	Single	7.0
	K	Single	18
	L	Single	29
	M	Single	109
Metric Power™ V-Belts Lengths ≤ 3000mm	XPZ	Single	51
	XPA	Single	87
	XPB	Single	156
	XPC	Single	249
	10X	Single	44
	13X	Single	82
	17X	Single	138

Ordering Guide



Please Specify

Media as:
Depth x
Width x
Height.

Dist. Pad as:
Thickness x
Depth x

Evaporative Media is priced in the following increments and is priced per cubic foot:
Depth (D): 4, 6, 8, 12, 18, 24"
Width (W): 12"
Height (H): 12, 18, 24, 30, 36, 48, 60, 72"
Heights exceeding 72" - made of stacked pieces.
Ex: 102" height = 1 piece 72" + 1 piece 30"

Distribution Pad is priced in the following increments and is priced per square foot:
Thickness (T): 2" (3" special order)
Depth (D): 4, 6, 8, 12, 18, 24"
Length (L): 12, 18, 24, 30, 36, 48, 60, 72"
Lengths exceeding 72" are made of multiple pieces.

Media cubic feet = $\frac{\text{unit depth" x unit width" x unit height"}}{1728}$

Distribution Pad square feet = $\frac{\text{depth" x length"}}{144}$

Example:
The above unit is measured as:
Media depth (D) = 12" Media width (W) = 54"
Media height (H) = 44"
Sizing:
Media depth (D) = 12": is a standard dimension
Media width (W) = 54" (next standard increment is 60")
The unit width will be made up of 4 pieces (E1, E2, E3, and E4) of 12" wide plus 1 piece (E5) of 12" wide cut to 6" wide.
Unit media height (H) = 44" (next standard increment 48")
All media is cut to the specified height of 44" at no additional charge.

Calculating Quantity for Pricing:
Media cubic feet = $\frac{12" \text{ d x } 60" \text{ w x } 48" \text{ h}}{1728}$ = 20 cu.ft.

Distribution Pad Square feet = $\frac{12"\text{d x } 60"\text{L}}{144}$ = 5 sq.ft.
(Note the 60"L piece of dist. Pad is cut to 54"L)

Master Industrial/Commercial HVAC Distributor:

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Distributed by:

Air Filter Solutions Inc.
2500 W. 8th Ave, Denver, CO 80204
Phone: (303) 893-3030 Fax: (303) 893-3031
info@airfiltersolutionsinc.com

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